

COLLIDING TRAVERSABLE WORMHOLES: GRAVITATIONAL WAVES AND THE FORMATION OF HEAVY BLACK HOLE SEEDS

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We present the first numerical-relativity evolutions of wormhole binaries: pairs of massive Ellis–Bronnikov drainholes held open by a ghost scalar field, evolved in 3+1 dimensions at equal masses, one throat compactness and zero spin. Each throat is an unstable fixed point whose e-fold time approaches the González–Guzmán–Sarbach linear rate with resolution, to 2.5% at our finest level, so its lifetime grows only as the logarithm of its initial perturbation. Two throats obey a charge-sign rule: like scalar charges repel and opposite charges attract, with a measured attraction-to-repulsion ratio of 1.462 ± 0.022 against the predicted $3/2$, so only opposite-signed pairs merge. They do not fuse into one wormhole. A head-on collision forms a common marginally outer trapped surface around both still-open throats, born with their combined area; its irreducible mass exceeds the ADM mass, so the Penrose inequality fails for this matter. The horizon then loses 28% of its area as it swallows the phantom field and settles near the Schwarzschild radius of the pair’s mass: a black hole with two wormholes inside it. An orbital merger ends in a remnant of the same size. Between 90% and 120% of the circular momentum the orbit turns from a scattering into a plunge; the scatterers form no horizon, radiate the most gravitational-wave energy—the output peaks at that boundary—and both mouths inflate. Gravitational waves are measured from both conversions. A lone collapsing throat sheds its seeded quadrupole as a burst ringing at the Schwarzschild frequency of the black hole it becomes—following the first gravitational-wave signal reported from a wormhole collapse, the massless throat’s [42]—and every encounter radiates its own burst of gravitational waves and a scalar dipole of comparable energy and negative sign, which a horizon switches off on the timescale on which the remnant’s mass settles. A throat survives at most a few orbital periods for any plausible perturbation, and the negative-energy dipole would pump rather than drain the orbit, so these binaries have no inspiral. Their signal is a single burst with no chirp before it; a search of 2.25 h of LIGO O3b data finds none. If such throats formed as late as $z \approx 20$, their collapse would supply halo-independent heavy black-hole seeds and millihertz bursts resolvable by LISA. The throats that inflate would create expanding anti-trapped regions from regular data, which the null energy condition otherwise forbids.

I. INTRODUCTION

Wheeler’s spacetime foam is a sea of handles—wormholes that open and close on the Planck scale [1–3]. Several mechanisms promote them to macroscopic size: inflation stretches a Planck-scale Lorentzian wormhole with the background [4]; bubble and domain-wall cosmologies produce post-inflationary regions joined through wormholes [5, 6, 89, 90]; and a gas of relic wormholes has been studied as a cosmology in its own right [7, 8]. The bubble and wall wormholes are not traversable and pinch off into black holes [6, 89]; none of this work evolves a traversable relic—what one does, and what two of them do to each other. That is the subject of this paper.

Two classical facts govern the dynamics. First, the simplest evolvable traversable wormhole—the Ellis–Bronnikov drainhole held open by a ghost scalar [9–12]—is unstable [15, 61, 85]: depending on the sign of the perturbation, the throat collapses to a black

hole or inflates, on a timescale set by the throat radius [13, 14, 73, 84, 86, 88], as 3+1 evolutions of the massless throat confirm nonlinearly [42]. As a quantum field coupled to gravity a ghost destabilises the vacuum unless its effective theory is cut off at low energy with Lorentz violation [71]; we treat it as a classical phenomenological model. Second, Geroch’s theorem forbids classical topology change [16, 17]: two wormholes cannot fuse into one, and none can be created classically. Creation is quantum [18], so our evolutions start at $t = 0$ from the newborn state—the stipulated initial data of every numerical binary, here with a physical origin attached.

Together these facts set selection rules. No astrophysical assembly channel exists: a throat decays faster than dynamical friction or radiation reaction could pair it. A primordial pair born from foam needs no assembly—birth and collision coincide, at a separation comparable to the objects—and its evolution is a race between orbital collision and throat decay, both rates fixed by the classical solution. The scalar adds a second rule: a phantom field repels like charges and attracts opposite ones, so like-signed pairs scatter while opposite-signed pairs fall together; and the sign of a lone throat’s perturbation selects collapse—a wormhole-to-primordial-black-hole channel beside the standard ones [19–21]—or infla-

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tion. A foam population thus splits into four fates—merger, scattering, collapse, inflation—across birth separation, perturbation and relative sign, the space mapped here. The binary evolved has three asymptotic ends: two independent wormholes, each leading to its own far universe. A foam handle differs—its two mouths belong to one wormhole and carry opposite scalar charge by construction—so our opposite-signed pairs reproduce a handle’s charges but not its topology. What merges is not two wormholes into one but two wormholes into one black hole containing both throats.

The observable is not a chirp but a population of bursts from the merger epoch, redshifted into whatever band the mass scale selects [22–25]. The epoch-independent deliverable is the energy radiated per event and its spectrum in throat-mass units (Sec. IX B). Every waveform behind it is extracted in the near zone, on spheres at $R = 10\text{--}44$, and none is extrapolated to null infinity: each radiated energy carries the spread across its spheres as a near-zone error bar (Sec. VIII). Collapse, alone or in a merger, also bears on JWST’s overmassive black holes at $z \gtrsim 4\text{--}6$, which favour heavy seeds of $10^4\text{--}10^6 M_\odot$ from formation channels that remain contested [44–47, 50]; Sec. X B explores what a population of such throats would imply for heavy seeds and millihertz bursts, conditional on the epoch at which they form, which our evolutions do not fix.

Sections II and III give the model and numerics; Secs. IV–VII follow the lone wormhole, two throats at rest, and the head-on and orbital mergers; Sec. VIII treats the radiation of every channel and Sec. IX confronts it with LIGO–Virgo–KAGRA data; Sec. X draws the consequences and Sec. XI gives reproducibility and cost. Appendix A holds the code-health record and Appendix B supplementary measurements. All evolutions use GRTeclyn [39], the AMReX-based [40] GPU port of GRChombo [41], continuing Refs. [42, 43]. We use $G = c = 1$.

II. THE MODEL

A. Field equations

The matter is a single real, massless phantom scalar: gravity couples to *minus* the canonical stress tensor,

$$T_{ab} = -\left(\nabla_a \phi \nabla_b \phi - \frac{1}{2} g_{ab} \nabla_c \phi \nabla^c \phi\right), \quad V(\phi) \equiv 0, \quad (1)$$

so the energy density seen by normal observers, $\rho = -\frac{1}{2}(\Pi^2 + \gamma^{ij} \partial_i \phi \partial_j \phi)$ with $\Pi = (\partial_t \phi - \beta^i \partial_i \phi)/\alpha$, is non-positive everywhere, and the null energy condition fails wherever ϕ varies—the exotic stress that holds a throat open. The field itself obeys the canonical wave equation $\square \phi = 0$, evolved in first-order form in ϕ and Π .

B. Drainhole initial data

Each body is a massive Ellis–Bronnikov drainhole [9, 10]. In isotropic radius r about the throat centre, with

$$X = \frac{r - a^2/4r}{a}, \quad \Omega = 1 + \frac{a^2}{4r^2}, \quad u = \frac{m}{a} \left(\arctan X - \frac{\pi}{2} \right), \quad (2)$$

the exact static solution of Eq. (1) is

$$\alpha = e^u, \quad \gamma_{ij} = e^{-2u} \Omega^2 \delta_{ij}, \quad \phi = C \arctan X, \quad (3)$$

with $4\pi a^2 C^2 = a^2 + m^2$, $K_{ij} = 0$, $\Pi = 0$ and zero shift; in the evolved conformal variables $\chi = e^{2u} \Omega^{-2}$ and $h_{ij} = \delta_{ij}$. Both constraints hold identically at $t = 0$. Here a is the throat scale and m the ADM mass, carried by the lapse rather than the conformal factor, so χ at the throat stays in $[0.15, 0.25]$ for $m/a \leq 1$ and mass costs no resolution. The minimal surface sits at $r_t = \frac{1}{2}(m + \sqrt{m^2 + a^2})$, where $X = m/a$, so its areal radius is analytic: $R_{\min} = e^{-u(r_t)} \sqrt{a^2 + m^2} = \sqrt{a^2 + m^2} \exp[(m/a) \arctan(a/m)]$, which is a at $m = 0$. The production throat $a = 2$, $m = 1$ has $R_{\min} = \sqrt{5} e^{\frac{1}{2} \arctan 2} = 3.8895$ at $r_t = \frac{1}{2}(1 + \sqrt{5}) = 1.618$. The point $r \rightarrow 0$ is the far side’s spatial infinity, compactified ($\chi \sim r^4$). At $m = 0$ the throat is ultrastatic ($\alpha \equiv 1$, $M_{\text{ADM}} = 0$) and two such throats do not attract.

C. Two throats in one slice

Two throats at separation d (one-body radii r_A , r_B) start from their combined fields,

$$u = u_A + u_B, \quad \psi = 1 + (\sqrt{\Omega_A} - 1) + (\sqrt{\Omega_B} - 1), \quad (4)$$

with $\chi = e^{2u} \psi^{-4}$, $\alpha = e^u$ and $\phi = \phi_A + \sigma \phi_B$, $\sigma = \pm 1$; the constant tail of the arctan profiles is subtracted so that $\phi \rightarrow 0$ at the boundary. This is the starting point of the constraint solve of Sec. II D, which turns it into initial data. Momentum is carried by the exact boost of Sec. II E: the throats sit at $x = \mp d/2$ with $P^i = (0, \pm p, 0)$, momenta of magnitude p in units of the one-body mass, tangential to the separation and opposite on the two throats, so the pair carries no net momentum and orbital angular momentum $J = pd$; $p = 0$ is a release from rest. The relative sign σ is the pair’s one discrete parameter. Far from its throat each body is a mass m carrying the scalar charge $q = Ca$ ($\phi_A \simeq -q/r_A$ once the constant tail is subtracted), and a phantom mediator *repels* like charges and *attracts* opposite ones, so two throats at rest a large distance d apart attract with

$$F = \frac{m^2}{d^2} (1 - \sigma Q), \quad Q \equiv \left| \frac{F_\phi}{F_{\text{grav}}} \right| = \frac{4\pi q^2}{m^2} = \frac{a^2 + m^2}{m^2} > 1 \quad (5)$$

always ($Q = 5$ for the production throat): a like-signed pair repels and never merges under its own gravity, and $\sigma = -1$ is the merger channel.

TABLE I. Each mouth of the matched $d = 8$ pair at $t = 0$ ($L = 64$, level 3) against the isolated throat: minimal areal radius, far-side mass [Eq. (7)], one-body mass m' , and the M_{ADM} of Eq. (8)—the throat’s own, and the pair’s.

data	R_{min}	M_{far}	m'	M_{ADM}
isolated throat	3.890	−4.810	1	1.001
matched pair, each mouth	3.892	−4.810	1	2.362

D. Constraint-solved data and matched placement

With $\Pi = 0$, $K = 0$ and $\gamma_{ij} = \Psi^4 \delta_{ij}$ the Hamiltonian constraint of Eq. (1) reads, in flat operators,

$$\nabla^2 \Psi - \pi |\nabla \phi|^2 \Psi + \frac{1}{8} \hat{A}_{ij} \hat{A}^{ij} \Psi^{-7} = 0. \quad (6)$$

The single drainhole solves it exactly. For a pair we solve it for a correction w to the starting Ψ on the whole initial mesh hierarchy with AMReX’s multigrid [40], holding ϕ , Π and the lapse fixed (a Robin outer boundary); a pair with momentum solves both constraints together (Sec. II E).

Which throat is each mouth? Near its centre $\Psi = c/r + d_c + O(r)$, and the inversion $r' = c^2/r$ turns this into an asymptotically flat end—the mouth’s own far universe—with ADM mass and scalar charge

$$M_{\text{far}} = 2c d_c, \quad Q_{\text{far}} = \frac{4C c^2}{a}, \quad (7)$$

both frozen in time, like a puncture’s individual mass. The isolated drainhole has $M_{\text{far}} = -m e^{\pi m/a} = -4.8105$ and $Q_{\text{far}} = \sqrt{a^2 + m^2} e^{\pi m/a} / \sqrt{4\pi} = 3.0344$. A pair $(M_{\text{far}}, Q_{\text{far}})$ names one isolated drainhole, and its near-side mass m' is the mouth’s one-body mass.

A second throat in the slice rescales each mouth: near A the companion’s constants multiply the throat’s own field, $\Psi \rightarrow e^{-u_B(x_A)/2} \Psi$, and because ϕ fixes a static throat’s coordinate size, the puncture coefficient c alone cannot undo this. The *matched placement* gives each throat a coordinate scale λ ($a \rightarrow \lambda a$ and $m \rightarrow \lambda m$ in ϕ , u and Ω ; C is scale-free), sets $\lambda = (c/c_{\text{iso}})^2$ with $c_{\text{iso}} = \frac{a}{2} e^{\pi m/2a}$, which holds Q_{far} at its isolated value, and iterates c until M_{far} is isolated too. Each mouth is then locally the isolated static drainhole (Table I): its R_{min} lands within 0.06% of R_* without being imposed, and $m' = 1$. With momentum the iteration converges the same way ($d = 12$, $p = 0.12$: R_{min} within 0.02% of R_*).

The pair’s mass. With $\Psi \rightarrow 1 + M_{\text{ADM}}/2r$, Gauss’s theorem on Eq. (6), whose punctures contribute $-4\pi c_X \delta_X$, gives

$$M_{\text{ADM}} = 2 \sum_X c_X - \frac{1}{2\pi} \int \left(\pi |\nabla \phi|^2 \Psi - \frac{1}{8} \hat{A}_{ij} \hat{A}^{ij} \Psi^{-7} \right) d^3x, \quad (8)$$

which returns the exact throat’s $M = 1$ to 0.14% and, unlike a fit to the monopole tail, does not read the outer

TABLE II. Mass of matched pairs at rest [Eq. (8)] against two isolated throats on the same grid ($L = 128$, $\Delta x = 0.0625$ at the throats), and Eq. (9). λ : each throat’s coordinate scale.

d	$\sigma = -1$ (attract)			$\sigma = +1$ (repel)		
	λ	E_b	Eq. (9)	λ	E_b	Eq. (9)
8	0.857	+0.357	+0.367	0.906	−0.619	−0.616
12	0.906	+0.269	+0.274	0.931	−0.437	−0.433
16	0.931	+0.214	+0.217	0.946	−0.338	−0.335
24	0.955	+0.151	+0.152	0.962	−0.233	−0.231
48	0.979	+0.080	+0.080	0.980	−0.121	−0.120

boundary’s offset of w as mass. For matched pairs at rest (Table II) the excess over two isolated throats follows

$$E_b \equiv M_{\text{ADM}} - 2m = -\lambda^2 \frac{m^2}{d} (1 + \sigma Q) \quad (9)$$

for both signs, to 3% at $d = 8$ and 0.5% at $d = 48$. It is the Newtonian binding $-\lambda^2 m^2/d$ plus the phantom cross energy $-\sigma \int \nabla \phi_A \cdot \nabla \phi_B d^3x = -\sigma \lambda^2 (a^2 + m^2)/d$ of two charges λq , with $\lambda \rightarrow 1$ as $d \rightarrow \infty$; the solve adds nothing else. In the merger channel $\sigma = -1$ the scalar term is positive and wins: the attracting pair weighs *more* than its parts, $E_b d \rightarrow +4$, and the repelling pair less, $E_b d \rightarrow -6$. Read at fixed charges, Eq. (9) would make the $\sigma = -1$ pair repel. Equation (5) follows instead if the scalar part acts at fixed potential, $F_\phi = +\partial E_\phi / \partial d$, as between conductors held at fixed voltage: the value of ϕ at each throat’s far infinity cannot change, while the flux through the throat can. The evolution decides for fixed potential: rest pairs on matched data pull and push in the ratio 1.462 ± 0.022 that Eq. (5) predicts (Sec. V A).

Units of this paper. Masses and times are in units of the drainhole parameter $m = 1$. Every production binary evolves matched, solved data: each mouth’s far side is the isolated throat’s, $m' = 1$ to 6×10^{-6} across the production set, so a quoted momentum p is per true one-body mass, with boost speed $v = p/\sqrt{1+p^2}$, and the pair’s mass is the solve’s M_{ADM} [Eq. (8); 2.3573 for the $d = 8$ head-on pair].

E. Throats with momentum

A throat with ADM momentum p is the exact drainhole boosted to $v = p/\sqrt{m^2 + p^2}$ and cut at $t = 0$: the metric, K_{ij} , ϕ and Π together, with the boosted lapse and shift, so that the slice is an exact solution of both constraints and the scalar moves with its throat ($\Pi \neq 0$). On the slice every χ contour, the throat’s included, is an ellipsoid contracted along the motion by $1/\gamma = m/\sqrt{m^2 + p^2}$, as a boosted black hole’s horizon is; the measured axis ratios follow this law to 6×10^{-5} at every p up to 0.45 (Fig. 1).

A pair is built from two boosted throats as in Sec. II D, with the companion’s K_{ij} and Π windowed out inside

each throat, and the solve then corrects both constraints, for w and a vector potential W ($\hat{A} \rightarrow \hat{A} + \mathbb{L}W$). Three checks anchor it: on a lone boosted throat the solve is idle ($\max |w| \leq 5 \times 10^{-7}$, $\max |W| \leq 5 \times 10^{-6}$, the far side isolated to 4×10^{-7}); on the rest pair it reproduces the scalar solve's c , σ and w_0 to 10^{-5} ; and on the flipped $d = 12$, $p = \pm 0.25$ pair on the production grid the far sides come out the isolated throat's to 10^{-5} , each mouth's $R_{\min}$ lands 0.09% from the isolated value, and the solved throat-shell Hamiltonian is at most 7.8×10^{-5} rms across the refinement levels. The pair's mass is the boosted background's own ADM energy—each throat's $\gamma\lambda_X m$ and its puncture shift, by the full surface integral of the closed forms—plus the monopole of w on the box faces, since Eq. (8) needs conformally flat, time-symmetric data; where both apply, the face value reads 2.8% below it.

F. Declared seeds

The instability is probed with declared perturbations of the conformal factor only: a Gaussian shell on the minimal surface,

$$\psi \rightarrow \psi[1 + \varepsilon g(r)], \quad g = e^{-(r-r_t)^2/w^2}, \quad w = a/4, \quad (10)$$

which changes the throat's areal radius by a fraction $\approx 2\varepsilon$, and a quadrupolar factor

$$\psi \rightarrow \psi[1 + \varepsilon_2 g(r) P_2(\cos \theta)] \quad (11)$$

about the z axis through the throat. Neither seed is constraint-solved. K_{ij} and Π stay untouched, so the momentum constraint holds at $t = 0$; but the drainhole solves the linear equation $\nabla^2 \Psi = \pi |\nabla \phi|^2 \Psi$ ($\Psi^4 = \chi^{-1}$, flat operators), so the spherical seed leaves exactly $\mathcal{H} = -8\varepsilon \Psi^{-4}(1 + \varepsilon g)^{-5}(\nabla^2 g + 2\nabla \ln \Psi \cdot \nabla g)$ —linear in ε , confined to the shell and, the shell being narrow, not small there: its peak is $0.93 \times 16\pi|\rho|$ at $|\varepsilon| = 10^{-2}$. The quadrupolar seed's defect is pure $\ell = 2$ at first order. CCZ4 feeds the defect into Θ and, within the first steps, into $\mathcal{M}$ (Fig. 10(b)). Above $\varepsilon \simeq 0.025$ the spherical seed turns the minimal surface into a maximum between two minima. The spherical seed selects the branch but cannot radiate: a spherical collapse has no $\ell \geq 2$ moment. The quadrupolar seed, with no $\ell = 0$ part at first order, leaves the branch to it and gives a collapsing throat a quadrupole to shed—a lone throat's only gravitational-wave signal (Sec. VIII A).

III. NUMERICAL SETUP

Table III groups the evolutions of the campaign.

A. Evolution scheme

We evolve CCZ4 [35] with damping $(\kappa_1, \kappa_2, \kappa_3) = (3, 0, 1)$, 1+log slicing [36] started from the exact static lapse of Eq. (3), a Gamma-driver shift [37, 38, 75] with $\eta = 1$ started from zero, fourth-order stencils, RK4 and Kreiss–Oliger dissipation $\sigma = 0.1$. The time step is $\Delta t = 0.02 \Delta x$, an order of magnitude below the usual CCZ4 Courant factor, because larger steps fail: a multiplier of 0.05 departs from the production run at $t \approx 33$, and 0.1 dies at $t = 16$. Every resolution statement below is at fixed $\Delta t/\Delta x$.

B. Grid, boundaries and regularisation

The domain is a cube of side $L = 64$ centred on the system, with a 128^3 base grid ($\Delta x = 0.5$). Nested boxes, halving per level and centred on the tracked throats (half-width 4 per throat at the production finest level), are re-drawn every 16 steps. Production runs use three refinement levels (finest $\Delta x = 1/16$); resolution arms go to levels 4–7 where stated, and every binary and two lone-throat arms use $L = 128$: the same cells in a box twice as wide. Level ℓ thus means a finest cell $\Delta x = 0.5/2^\ell$. The one exception is the octant of an $L = 512$ box that carries the inflating throat (Sec. IV F): its base spacing is 2, so its 5 levels give $\Delta x = 1/16$, the resolution of our level-3 arms, and we quote it by that Δx . Outer boundaries are Sommerfeld, backed by a spherical sponge about the box centre over the outer quarter of the half-width— $r = 24$ –32 at $L = 64$ and 48–64 at $L = 128$ —in which extra dissipation ramps to $\sigma_{\text{sp}} = 4$ as the fourth power of the depth; every extraction sphere lies inside its inner radius. The sponge is Kreiss–Oliger dissipation only: it damps grid-scale noise but is transparent to long-wavelength gauge and physical waves, which reflect from the box faces (Sec. VIB). The vacuum controls (Sec. VIIE) instead run GRTeclyn's binary-black-hole example with its own numerical settings and no sponge.

Each throat carries a compactified second infinity at its centre, where $\chi \sim r^4$, so the evolved variables are floored at $\chi \geq 10^{-8}$ and $\alpha \geq 10^{-10}$ (a variant floors χ at 10^{-20} and applies 10^{-8} only where $1/\chi$ is evaluated); χ -floor twins at 10^{-8} and 10^{-20} re-converge to four digits after touching the floor. Every boosted evolution except the $d = 12$, $p = 0.12$ arm switches the 1+log source off within coordinate radius 0.3 of each tracked throat centre, the far side's compactified infinity, restoring it by 0.8 (the throat sits at $r_t = 1.618$), so that the moving puncture does not drive the slicing. The $d = 6$ chain and the tangential $d = 12$ arms at $p \geq 0.25$ damp the scalar exponentially wherever the lapse has collapsed below 3×10^{-2} : inside the merged cores, and between the scatterers' mouths after the approach.

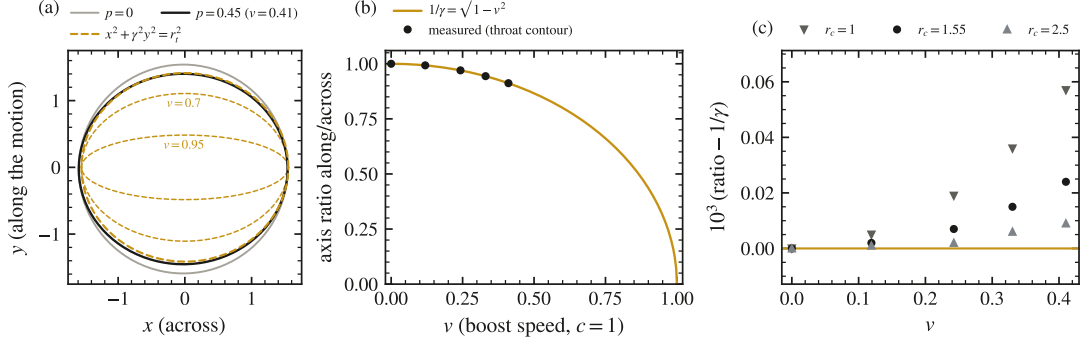


FIG. 1. The moving throat at $t = 0$ is the Lorentz-contracted drainhole. (a) χ contours of the throat at rest and at $p = 0.45$, the latter on its analytic ellipse $x^2 + \gamma^2 y^2 = r_t^2$; the fainter ellipses continue the law to $v = 0.7$ and 0.95 . (b) Axis ratio (along/across the motion) against the boost speed $v = p/\sqrt{m^2 + p^2}$, on $1/\gamma = \sqrt{1 - v^2}$ drawn to the disc limit $v = 1$. (c) The residual: below 6×10^{-5} at every v and at three contour levels. The outcome: the boosted data carry exactly the special-relativistic shape, with the scalar in motion, and both constraints hold at the exact solution’s discretisation floor.

TABLE III. The run matrix: the 96 packed evolutions of the campaign—constraint-solved initial data, momentum carried by the exact Lorentz boost (Sec. II E)—in groups, each varying one knob against a named reference or repeating one; per-run lines are in the shipped registry. Left: the single throat and the initial-data machinery. Right: the binaries and their controls. Unless stated, $a = 2$, $m = 1$ ($R_{\min} = 3.89$), $\sigma = -1$, $d = 12$ and $L = 64$.

group	knob(s)	runs	Sec.	group	knob(s)	runs	Sec.
lone throat	levels 2–4; χ floor; Δt ; $L=128$	10	IV	two throats at rest	σ ; $a = 1$ –3; $d = 12$ –18	8	V
spherical kicks	$\varepsilon = \pm 10^{-3} \dots \pm 10^{-1}$; levels 3–5	10	IV C	head-on, $d=8$	legs, levels 4–6; MOTS re-play; first-law window	8	VI
quadrupolar kicks	$\varepsilon_2 = 0.005$ – 0.05 ; kick-free arm	6	VIII A	merger chain, $d=6$	$p=0.10$ tangential; levels 3–7; σ_{KO} ; χ floor; settle leg	7	VIII C
scalar-stream re-runs	unkicked level 4, $L=64/128$; $\varepsilon=+10^{-2}$	3	VIII G	$d=12$ arms	$p = 0.12$ tangential, 0.25 inward	2	VIII B
boosted throat	$p = 0$ – 0.45 at $t=0$; collapse twins, levels 3/4	7	II E	fly-by and E_{GW} scan	$p = 0.25$ – 0.90 ; χ -floor and level-5 extensions	8	VIII D
solve verifications	boosted pairs at $t=0$ on the production grid; the pair at level 3 to $t=20$; a kicked single through the solve	4	II E	vacuum BBH	same d , p , masses; no scalar, no sponge	5	VIII E
placement	$d = 6$ – 48 , one step each	18	II D				

C. Diagnostics

Every coarse step the code writes the volume-averaged norms $\|\mathcal{H}\| = (\int \mathcal{H}^2 dV/L^3)^{1/2}$ and $\|\mathcal{M}\| = (\int \mathcal{M}_i \mathcal{M}_i dV/L^3)^{1/2}$, evaluated with fourth-order differences on the base grid over the whole domain after the finer levels are restricted onto it: they do not resolve the throats, monitor code health rather than local accuracy, and compare only within one box. Both throat centres are tracked, and about each the areal radius $R(r)$ is scanned on coordinate spheres out to $r = 2.79$; its minimum is the throat, and an inflating throat that outgrows the scan is bounded from above by the edge reading. On the same spheres the null expansions $\theta_{\pm}$ are computed with the outward direction fixed by *increasing areal radius*, so that a dynamic throat is marginally *anti-trapped* [33, 73]; the in-code monitor orients by co-

ordinate radius, which points the wrong way beyond a throat, and serves only as a live monitor.

The star-shaped scan reports the outermost coordinate sphere that is trapped everywhere ($\max \theta_+ = 0$, $\theta_- < 0$). A trapped sphere lies inside the apparent horizon, so this sphere—“the scan’s MOTS” below—bounds the marginally outer trapped surface from inside. On its spheres, as on the finder’s surfaces of Sec. III D, we evaluate the areal radius R and the Misner–Sharp mass [32],

$$\frac{2M_{\text{MS}}}{R} = 1 + \frac{R^2 \langle \theta_+ \theta_- \rangle}{4}, \quad (12)$$

with the product averaged over the sphere and $\theta_{\pm}$ computed for the null normals $k^a = n^a + s^a$ and $l^a = n^a - s^a$ (n^a the slice’s unit normal, s^a the sphere’s outward unit normal, so $k \cdot l = -2$); with the normalisation $k \cdot l = -1$ the factor $1/4$ becomes $1/2$. On an exact MOTS $M_{\text{MS}} = R/2$; on the scan’s spheres M_{MS} exceeds

$R/2$ by at most 3% about a lone collapsed throat, which stays nearly round.

Radiation is read from Ψ_4 on spheres at $R = 10, 14, 18$ (and 22 on the single-throat wave arms) on the single throats, and at $R = 20, 28, 36, 44$ on every binary, the head-on adding the single throats' three, decomposed in spin-weight -2 harmonics about z , the orbital axis. The head-ons fall along x : there the z -based $(2, 0)$ is $-h'_{20}/2$ of the quadrupole h'_{20} about the collision axis and carries a quarter of its power, the $(2, \pm 2)$ the rest ($|h_{22}/h_{20}| = \sqrt{3/2}$ to 2% on every sphere of both head-ons), so their $(2, 0)$ is quoted as h'_{20} . Every amplitude quoted is $r\Psi_4$; on the head-on the in-code and offline pipelines agree at the burst peak to 0.2%, while on the single throats the in-code stream is dominated by its grid-scale floor and only the offline one is used. A scalar stream is read on the single throats' spheres and, on the binaries, at $R = 14, 20, 30$ and 44 (the head-on adding 10 and 18), where the offline pipeline also reads Ψ_4 : ϕ and Π in spin-weight-0 harmonics, $A_{\ell m} = R\phi_{\ell m}$, and the kinematic flux $F_{\text{kin}} = -R^2 \oint \Pi \partial_r \phi d\Omega$, positive for an outgoing *canonical* wave (Sec. VIII F). A per-cell autopsy records the location and one-step history of any floating-point failure.

D. Marginally outer trapped surfaces

A marginally outer trapped surface (MOTS) is a closed two-surface of the slice on which the outgoing null expansion vanishes, $\theta_+ = 0$, while the ingoing one is negative, $\theta_- < 0$; the outermost MOTS is the apparent horizon. A MOTS is read off a single slice, unlike the event horizon, and since the phantom violates the null energy condition it need not lie inside one: every “horizon”, “censored” and “black hole” in this paper is a statement about MOTSs on the evolved slices.

The finder runs in post-processing on every archived slice, so horizon coverage matches the plot cadence. It expands the surface about the tracked centre as $r = h(\theta, \phi)$ in real spherical harmonics up to degree $\ell_{\text{max}} = 6$ on the head-on slices and $\ell_{\text{max}} = 8$ on the orbital ones, and solves $\theta_+[h] = 0$ by Newton iteration on metric and extrinsic-curvature data interpolated from the finest refinement level covering the surface, with the areal orientation of Sec. III C. The first surface of a run is found by the damped fast flow [74]; each later slice warm-starts from the previous surface and converges in a few iterations to a root-mean-square residual below the solver tolerance of 10^{-6} . From the converged surface we quote the areal radius $R = \sqrt{A/4\pi}$, the Misner–Sharp mass of Eq. (12), and the harmonic content beyond $\ell = 0$ as the deformation. The instrument recovers Kerr–Schild Schwarzschild ($R = 2M$ to 0.24%, $M_{\text{MS}} = M$ to 0.12%) and finds no trapped surface outside the time-symmetric Ellis throat. Every binary horizon number in this paper—formation times, R , M_{MS} —is read from these surfaces, the lone throats' nearly round collapse horizons from the star-shaped scan (within 3%, Sec. III C). On a binary

the scan is only an inner bound and a live monitor: a fully trapped round sphere lies strictly inside a deformed MOTS, so the scan under-reads a forming horizon, or misses it.

IV. THE SINGLE WORMHOLE

A. The static throat as a fixed point

Evolved without a declared perturbation, the exact drainhole holds $R_\star = 3.8895$ to 0.1% for 35 time units at level 3 and to 1% until $t \simeq 44$ (Fig. 2(a)); at level 4 it is exact to four digits at $t = 30$. The departure moves *later* by 5.6–8.7 units per halving of the cell—a truncation-noise seed of fixed growth rate handed a smaller amplitude. Level 2 dies of an origin failure at $t = 24.2$ with the throat still exact to 0.12%. A lighter throat is less stable: at level 3 the $m = 0.5$ throat holds 0.1% only to $t \approx 17$, against 35 at $m = 1$, and dies at its origin at $t = 26.3$.

B. The unstable mode

The departure is a single exponential mode (Fig. 2(a,b)): plateau fits give $\tau = 5.88$ (level 3) and 5.26 (level 4), i.e. $T \equiv \tau \alpha_{\text{th}}/R_\star = 0.87$ and 0.78, where $\alpha_{\text{th}} = e^{u(r_\star)} = 0.575$ is the static lapse at the throat, so that T is the proper-time e-fold in units of the throat's light-crossing time. Our drainhole is the $\gamma_1 = m/a = 0.5$ member of the static family of González, Guzmán and Sarbach (their metric with $b = a$, $\gamma_0 = -\pi\gamma_1/2$), for which their linear analysis gives $T = 0.758$; their Table I spans $T = 0.846$ at the massless end to 0.590 [15]. Solving the linearised field equations of this solution ourselves—their regularised master equation and, independently, the reduction in the gauge $\delta\phi = 0$ —gives $T = 0.7584$, i.e. $\tau_{\text{lin}} = 5.13 M$. The measured e-fold approaches it with resolution: it is 2.5% longer at level 4 and 15% longer at level 3; with two resolutions we quote no convergence order. The seed is truncation error, and its sign flips with resolution—level 3 collapses, level 4 inflates—confirming from exact massive data the two branches found for the massless throat [42].

C. Branching: collapse versus inflation

A declared kick, Eq. (10), overrides the truncation seed (Fig. 2(c)). At $\varepsilon = \pm 10^{-2}$ and $\pm 10^{-3}$ the fate is always opposite to the push, each $\pm$ pair crosses at the ε -independent $t_\times \simeq 13.0$, and a tenfold smaller kick arrives 11–12 units later at matched horizon radius—linear response on the same mode. The branching survives refinement: at level 4, whose own seed inflates the un-kicked throat, $+10^{-2}$ still forms a horizon at $t = 11$ and -10^{-2} still inflates, the twins crossing at $t = 12.99$ against level

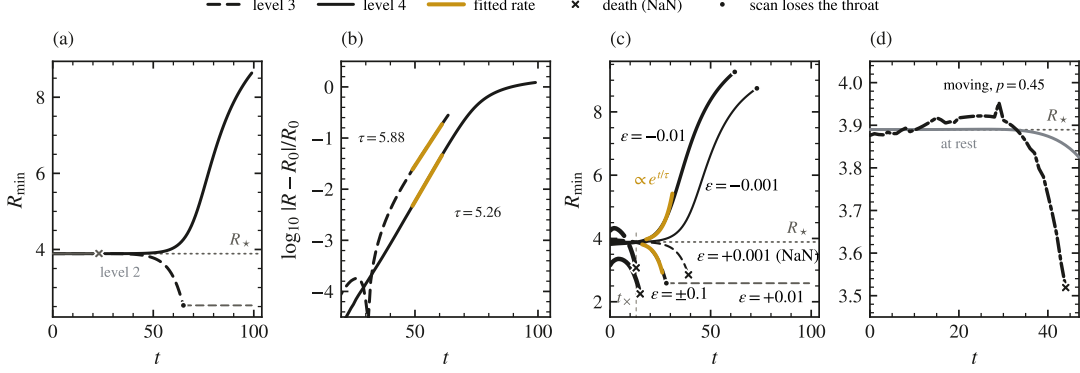


FIG. 2. Instability of the isolated throat. (a) Minimal areal radius from exact static data at refinement levels 2–4. The exact solution (R_* , dotted) is held until truncation noise excites the unstable mode: level 3 (dashed) collapses, level 4 (solid) inflates, level 2 (grey) dies of an origin failure first (cross). Dots mark where the scan loses the throat; muted lines carry the last reading. (b) Deviation from R_* : one exponential mode with a level-dependent sign, $\tau = 5.88/5.26$ at levels 3/4. (c) Level 3 with declared seeds, Eq. (10): $\epsilon = \pm 10^{-2}$ (heavy), $\pm 10^{-3}$ (light) and, heaviest, ± 0.1 , which both collapse; dashed, outward kicks. Twins cross at $t_{\times} \simeq 13.0$ independent of ϵ . Gold: exponential fits; crosses: deaths at the compactified origin, $t = 40.1$ ($+10^{-3}$), 14.1 ($+0.1$) and 15.2 (-0.1). Their constraint norms are Fig. 10(a,b). Unless a seed is declared, the grid’s truncation noise picks the branch. (d) The throat in motion (Sec. IV D): the exact Lorentz-boosted throat (dash-dot, $p = 0.45$, level 3) against the resting level-3 arm (grey). It rides R_* to $t \approx 29$ and then collapses on the level-3 mode, dying at $t = 44.67$: the fate and its rate are the resting throat’s, carried with the motion.

3’s 13.01. The branch point is the throat’s, not the grid’s or the kick’s. The kick also sets the clock: horizons form at $t = 11$ ($\epsilon = +10^{-2}$) and $t = 25$ ($+10^{-3}$), against $t \leq 61$ for the unkicked throat (its first scanned slice after departure already holds a MOTS). At $\epsilon = \pm 0.1$ the data are no longer perturbative (Fig. 2(c)): $+0.1$ turns the minimal surface into a maximum between two minima and is trapped from the first scan ($t = 1$), while -0.1 , which narrows the throat to $R_{\min} = 3.15$, re-expands to 3.36 by $t = 4$ and then collapses as well, trapped from $t = 6$ —against the sign rule. Both die at the compactified origin ($t = 14.1$ and 15.2 , NaN in h_{11}), the failure that also ends the $+10^{-3}$ arm at $t = 40.1$, before the lapse has collapsed anywhere. Their Hamiltonian defect is $3\text{--}6\times$ that of the 1% arms in the box norm and $8.6\text{--}10.3 \times 16\pi|\rho|$ on the shell; the usable range ends at 10^{-2} .

D. The throat in motion

A uniformly moving throat keeps the resting throat’s fate. The exact-boost $p = 0.45$ throat (Sec. II E; level 3, $v = 0.41$) rides R_* , rising at most 2.0%, until $t = 36.8$, where it falls 1% below its start (the resting level-3 throat: $t \simeq 44$), and then collapses on the resting throat’s own mode: $\tau = 5.5\text{--}5.7$ against 5.88 at rest, on the same branch—level 3 collapses—and it dies at the same numerical endpoint ($t = 44.67$, Fig. 2(a)). The instability is the throat’s, carried with the motion; the momentum changes neither the branch nor the rate.

E. The collapse endpoint

The collapse branch ends in a black hole that consumes its supporting field. On the unkicked level-3 arm the horizon’s areal radius falls $3.23 \rightarrow 2.37$ and $M_{\text{MS}} 1.62 \rightarrow 1.23$ over 39 units, with the origin lapse at 0.016; the kicked $\epsilon = +10^{-2}$ arm runs clean to $t = 100$ behind its horizon. Unlike the massless throat of Ref. [42], which bounced at $t \approx 4M$ and shed its horizon by $18.5M$, no arm here sheds a horizon. That collapse was driven with the phantom’s stress-energy halved in the Einstein equations; the unmodified evolutions here supersede it where they differ. The pure-quadrupole arm ($\epsilon_2 = 10^{-2}$, no radial kick) is fully instrumented (Fig. 3): a MOTS from $t = 33$ at $R = 3.80$, shrinking to 2.41 ($M_{\text{MS}} 1.90 \rightarrow 1.24$), while $|K|$ grows inward and the lapse pinches to zero at the centre.

The shrinking is physical. In spherical symmetry a massless phantom can only shrink a horizon: with the double-null metric $-2e^{-f}du dv + R^2 d\Omega^2$, $T_{uv} = 0$ and $T_{vv} = -(\partial_v \phi)^2$, the Misner–Sharp mass on a marginally trapped tube obeys $\partial_v m = 4\pi R^2 e^f (\partial_v \phi)^2 \partial_u R \leq 0$ and $\partial_u m = 0$, and the tube is timelike or null [62, 72], so $R = 2m$ can only fall, as phantom accretion requires [63, 83]. The arms kicked with $\epsilon = +10^{-2}$ shrink by 40% in areal radius to a floor at $t \approx 47$; the regrowth their scans show after it tracks the unconstrained seed’s Hamiltonian defect (Sec. II F) reaching the horizon, and is absent from the pure-quadrupole arm, whose seed has no $\ell = 0$ defect (Fig. 3(a)). The throat becomes a black hole that swallows its own phantom support and loses 40% of its horizon radius doing so, in minutes even at $10^6 M_\odot$ (Table IV): Hawking’s area law assumes the null

energy condition [34, 67], and this horizon runs it backwards.

F. The inflation branch

The inflation branch forms no trapped surface. The $\varepsilon = -10^{-2}$ throat, evolved on an octant of an $L = 512$ box with finest $\Delta x = 1/16$ (the resolution of our $L = 64$ level-3 arms; Fig. 3), keeps growing to the end of the record, $\times 3.8$ by $t = 218$, and stays anti-trapped [73]: the $\theta_l = 0$ horizon sits on the neck and $\theta_k = 0$ runs outward, to $R \approx 73$ by $t = 212$. Until $t \approx 40$ the neck grows exponentially in its own proper time with $HR_0 = 1.22$ (locally 1.0–1.3; $R_0 = 3.81$, the kicked neck), 6% below our linear mode’s 1.30, consistent with this arm’s level-3-class resolution. Spherical evolutions find the same proper-time exponential: $Ha \approx 1.1$ for the massless throat [13] and exponential growth for massive ones [14]. After that the 1+log slicing freezes the lapse at the neck ($0.57 \rightarrow 0.015$), only ≈ 5 units of its proper time pass by $t = 218$, and its growth per unit t slows—a property of the slicing, not of the throat.

The throat is shedding its own support: the monopole scalar flux through every sphere grows quasi-exponentially over the quotable window (Sec. VIII G), drawing on a store that outside the static throat is a Misner–Sharp deficit of $R_*/2 - m = 0.94$; while the exterior slicing is still quiet the arms of Sec. VIII G shed 0.03 of it through $R = 18$, a few percent. By $t = 218$ neither a downturn of $R_{\min}(t)$ nor a trapped surface occurs: the growth holds at every sample, and the $\theta_k = 0$ surface keeps moving outward with no slowing over the last fifty units.

G. Lifetime and physical scale

At fixed m/a the solution is scale-free, and the measured e-fold is $\tau\alpha_{\text{th}} = 0.78\text{--}0.87 R_*$ in proper time—one light-crossing of the throat. The lifetime is a logarithm of the seed: the horizon times of Sec. IV C set a decade’s cost at $14M \simeq \tau \ln 10$, and the unkicked throat’s places the level-3 truncation seed at $\varepsilon \gtrsim 2 \times 10^{-6}$. Table IV converts these to physical units. No static throat of astrophysical mass survives: a stellar-mass drainhole is gone in milliseconds, a supermassive one in minutes, and only at $10^9 M_\odot$ does the lifetime reach days. The logarithm makes this robust: ten further decades of quiet initial data extend a $30 M_\odot$ throat from 1.6 to 22 ms. A foam-born pair must collide as it is born; otherwise each throat collapses or inflates on its own first (Sec. X A).

H. A type-I critical solution

In the language of critical collapse the static throat is a type-I critical solution [93]: a static solution with a sin-

TABLE IV. The drainhole clock in physical units: the production throat ($R_* = 3.89M$) scaled by its ADM mass M . $\tau = 5.9M$ is the e-fold time (Sec. IV B); the last three columns are the measured times to a horizon for the two declared kicks, both outward ($\varepsilon > 0$, the collapsing sign; Fig. 2(c)), and an upper bound for truncation noise (Sec. IV C). Intermediate seeds follow $t_{\text{BH}} \simeq 11M + \tau \ln(10^{-2}/\varepsilon)$.

M	R_*	τ	time to a horizon		
			$\varepsilon = +10^{-2}$	$+10^{-3}$	noise ($\leq$)
$30 M_\odot$	172 km	0.87 ms	1.6 ms	3.7 ms	9.0 ms
$10^3 M_\odot$	5.7×10^3 km	29 ms	54 ms	0.12 s	0.30 s
$10^6 M_\odot$	5.7×10^6 km	29 s	54 s	2.1 min	5.0 min
$10^9 M_\odot$	38 AU	8.1 h	15 h	1.4 d	3.5 d

gle unstable mode (Sec. IV B), sitting on the threshold between two end states—collapse and inflation, where the standard problem has collapse and dispersal. The measurements above carry both type-I signatures. The first is time scaling: data a distance ε from threshold stay near the critical solution for a time $\tau \ln(1/\varepsilon)$ set by its one e-fold, which is the lifetime law of Table IV, $t_{\text{BH}} \simeq 11M + \tau \ln(10^{-2}/\varepsilon)$, and both branches leave along the same mode, each $\pm\varepsilon$ pair crossing at the same $t_\times$ (Sec. IV C). The second is a mass gap: black-hole formation turns on at finite mass, as at the static Bartnik–McKinnon solution of Yang–Mills collapse [94], not at the vanishing mass of type-II scaling. The $\varepsilon = +10^{-2}$ and $+10^{-3}$ horizons form at $R = 3.88$ and 3.81 , both close to the throat’s own R_* : a tenfold smaller kick leaves the black hole’s mass at formation almost unchanged, where type-II scaling would shrink it as a power of ε . Shinkai and Hayward found evidence, in spherical evolutions of the massless throat, for critical solutions with a definite black-hole mass [13]; here the critical solution is the static throat itself, its one unstable mode measured in 3+1 dimensions.

V. TWO THROATS AT REST

A. The sign rule

Two identical throats released from rest push apart; reversing one throat’s scalar, $\sigma = +1 \rightarrow -1$ in Eq. (5), turns the push into a pull (Fig. 4(a,b)). Both forces share one radial law, so their ratio Q is separation-independent: no gap, mass or approach speed lets a like-signed pair fall together, and $\sigma = -1$ is the only gravity-driven merger channel. The flip is not a tuning: $\phi \rightarrow -\phi$ leaves T_{ab} , and with it the one-body geometry, unchanged. The measured pull-to-push ratio at $a = 2$ is 1.462 ± 0.022 (scatter over 15 slices) against the point-charge, large-separation $(Q+1)/(Q-1) = 3/2$ of Eq. (5), and the mass is equal in all arms, so the asymmetry is the scalar charge’s.

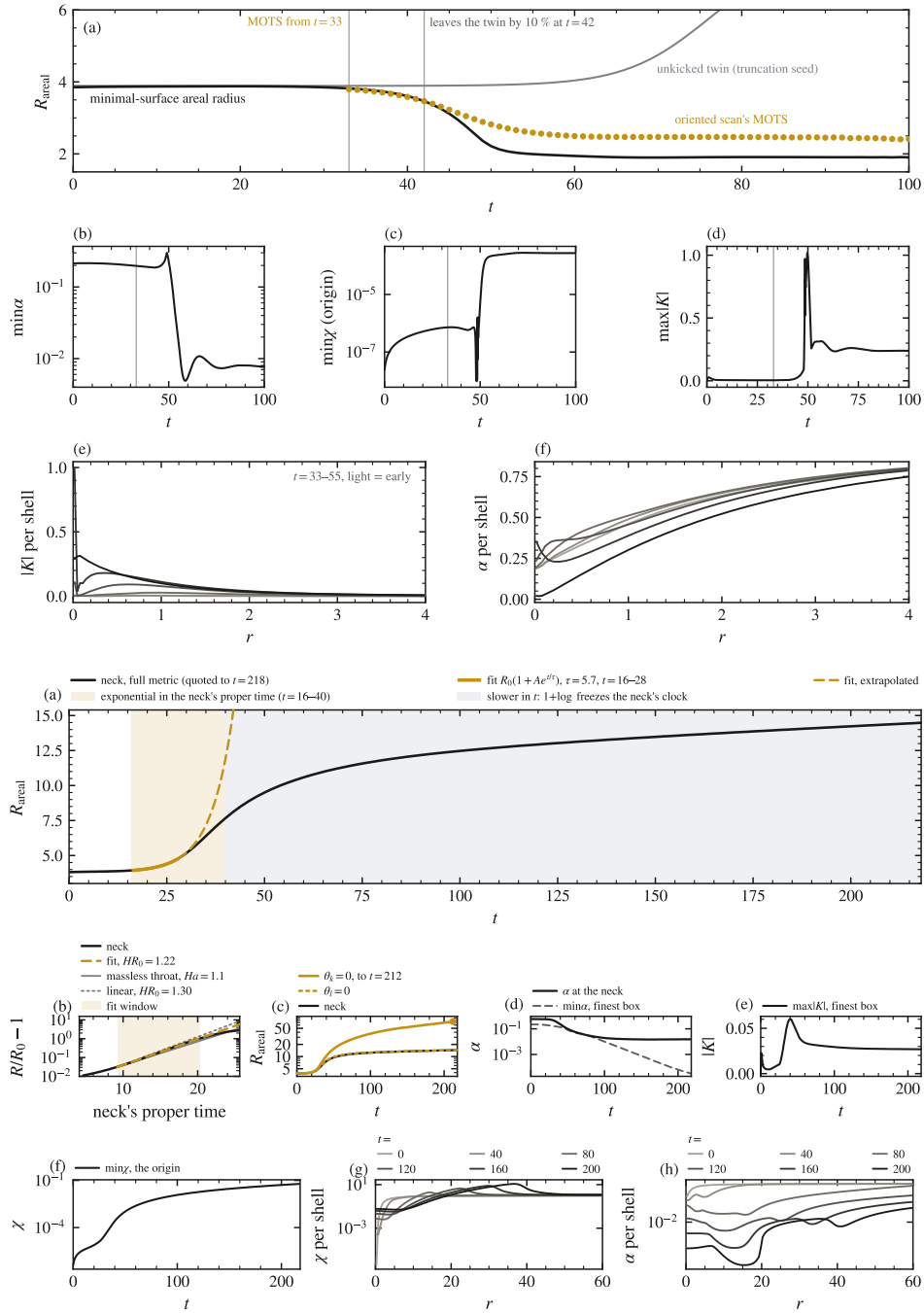


FIG. 3. *Top (collapse branch):* Collapse of the pure-quadrupole arm ($\varepsilon_2 = 0.01$, no radial kick). (a) Minimal-surface areal radius (ink) and the scan's MOTS (gold) from $t = 33$: R 3.80 $\rightarrow$ 2.41, M_{MS} 1.90 $\rightarrow$ 1.24. Grey: the unkicked level-4 twin, which inflates on its own seed; this arm leaves it by 10% only at $t = 42$ —the second-order drive of the pure quadrupole. (b)–(d) Global $\min \alpha$, $\min \chi$ (the origin, the far side's compactified infinity) and $\max |K|$; the violent phase spans $t = 45$ –55. (e),(f) Radial profiles over 128 shells, $t = 33$ –55 (light = early): $|K|$ grows inward while α pinches to zero at the centre. Constraint norms: Fig. 10(c). The collapse ends in a black hole that consumes its supporting field. Movie: Video 1; youtu.be/Cc87mHcBPOA. *Bottom (inflation branch):* The inflating throat ($\varepsilon = -10^{-2}$; $L = 512$ octant, finest $\Delta x = 1/16$) to $t = 218$. (a) Neck areal radius in the full metric, 3.81 $\rightarrow$ 14.5; gold, the exponential onset, $\tau = 5.7$ over $t = 16$ –28; shaded, exponential growth in the neck's proper time ($t = 16$ –40, gold) and the slower growth in t once 1+log freezes the neck's clock (grey). (b) Over that window the neck's proper-time rate is $HR_0 = 1.22$ (local 1.0–1.3), against the massless throat's $Ha = 1.1$ [13] and the linear mode's 1.30. (c) The $\theta_l = 0$ horizon stays on the neck; $\theta_k = 0$ runs out to $R \approx 73$ by $t = 212$. (d) The lapse at the neck falls 0.57 $\rightarrow$ 0.015; dashed, its minimum on the finest box. (e),(f) $\max |K|$ and $\min \chi$ on the finest box, around the origin. (g),(h) χ and α per shell (light = early): the χ peak moves out with the neck. Constraint norms: Fig. 10(d). The throat keeps growing, anti-trapped; its onset is at the linear rate. Movie: Video 2; youtu.be/RIQCitu4tSc.

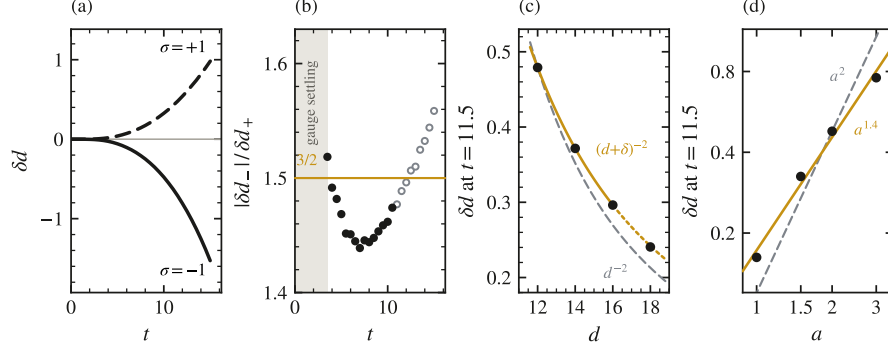


FIG. 4. Two throats at rest, on the far-side-matched pairs (Sec. IID): the sign rule (a,b), the force law (c) and its growth with throat width (d). (a) Separation change of two matched $d = 12$ pairs one sign apart: like-signed ($\sigma = +1$, dashed) opens, opposite-signed ($\sigma = -1$, solid) closes. Separations are inverse- χ -weighted centroids of the midplane χ pits. (b) Pull-to-push ratio against the prediction $(Q+1)/(Q-1) = 3/2$ (gold), where $Q = |F_\phi/F_{\text{grav}}| = (a^2 + m^2)/m^2 = 5$ is the scalar-to-gravity force ratio of Eq. (5): measured 1.462 ± 0.022 (scatter over 15 slices); grey, the gauge-settling window $t < 3.5$, excluded; open circles past $t = 10.5$, where the closing pair has eaten a tenth of its gap and the displacements are no longer small, excluded too. (c) The force law on the ladder: how far like-signed pairs released from rest at $d = 12/14/16/18$ have pushed themselves apart by $t = 11.5$ (dots), against inverse-square in the coordinate separation scaled through $d = 12$ (grey, dashed) and $A/(d + \delta)^2$ fitted on $d = 12\text{--}16$ alone (gold), dotted where it predicts the $d = 18$ rung (Sec. VB). If the push were inverse-square in d , every dot would lie on the grey curve; the dots rise above it, and the gold fit absorbs the excess with the offset δ . (d) The push against throat width, on logarithmic axes: how far like-signed $d = 12$ pairs of width $a = 1/1.5/2/3$ have pushed themselves apart by $t = 11.5$ (dots; $a = 2$ is the $d = 12$ rung of (c)), against the point-charge a^2 of Eq. (5) scaled through $a = 2$ (grey, dashed) and a power law fitted to all four rungs (gold). If the push grew as a^2 , every dot would lie on the grey line; the dots climb more slowly, as $a^{1.4}$.

B. The force law

How the push falls with distance is read off a ladder (Fig. 4(c)). Four like-signed pairs, identical but for the gap $d = 12/14/16/18$, are released from rest, and by a common $t = 11.5$ their separations have grown by $\delta d = 0.4791/0.3716/0.2963/0.2406$. For displacements this small, δd at a fixed time is proportional to the force, so under the inverse-square law of Eq. (5) $\delta d d^2$ would be the same on every rung. It is not: it rises monotonically, $69.0 \rightarrow 78.0$, across the ladder, which excludes inverse-square in the coordinate separation. Instead $F \propto (d + \delta)^{-2}$ with $\delta \approx 2.2\text{--}2.8$ describes it: δ fitted on $d = 12\text{--}16$ alone predicts 0.242 at $d = 18$, where inverse-square gives 0.213 and the measurement is 0.2406; the full-ladder fit gives $\delta = 2.65$. Since δ is a sizable fraction of the mouth radius $R_\star = 3.89$, the force is inverse-square in an *effective* separation exceeding the coordinate one by most of a throat radius. Across widths, at $d = 12$, the push grows at every rung (Fig. 4(d)): $\delta d = 0.1623/0.3250/0.4791/0.7583$ by $t = 11.5$ for $a = 1/1.5/2/3$, a power law $\delta d \propto a^n$ with $n = 1.2\text{--}1.4$ when fitted at $t = 8\text{--}11.5$, where point charges, whose net push $(Q-1)m^2/d^2$ is a^2/d^2 by Eq. (5), give $n = 2$.

VI. THE HEAD-ON MERGER

A. Contact and the common horizon

Released from rest— $p = 0$ —at $d = 8$ with opposite scalar signs, on matched, solved data ($M_{\text{ADM}} = 2.3573$), the pair falls together, and the spectral finder (Sec. IIID) holds a common MOTS around both mouths from $t = 18$ (Fig. 5): born while the χ pits are still ~ 3.4 coordinate units apart, a tenth out of round (deformation 0.104), with $R = 5.634$, $M_{\text{MS}} = 2.817$ and 1.05 times the two throats' summed area. No surface ever closes about either mouth alone: the horizon is born common. At birth the Penrose inequality is violated outright— $R/2 = M_{\text{MS}}$ exceeds M_{ADM} by 19.5%—as the phantom permits, since the inequality presupposes the non-negative energy density it lacks [76, 77]: each throat's own Misner–Sharp mass, $R_\star/2 = 1.94$, already exceeds its ADM mass of 1 by the negative energy of the field outside it. From birth the horizon only shrinks (Sec. VIB). The round star scan, the live monitor, finds the birth 4 units late and 11% small and goes blind while the surface is deformed (Fig. 5(a)): the scan-versus-MOTS offset of Sec. IIIC.

B. Past the interior failure to a black hole

The merged core is the grid's hard part, not the space-time's: 21 units after the horizon forms the level-5 leg dies of the censored single-cell NaN of Sec. VIID ($t = 38.845$),

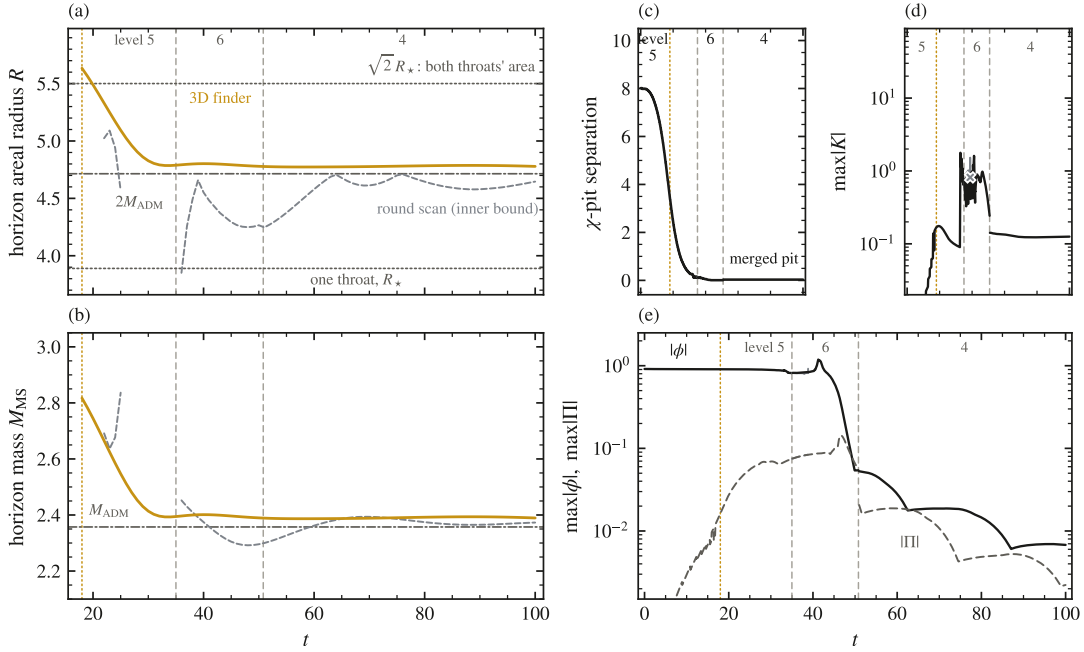


FIG. 5. The head-on merger ($d = 8$ from rest, $p = 0$, opposite signs) on matched, solved data ($M_{\text{ADM}} = 2.3573$), and the black hole it makes. One evolution in three legs (faint dashed rules at the seams, $t = 35$ and 50.8 ; levels named in each panel; Sec. VIB): level 5 to a NaN in h_{11} at the merged core at $t = 38.845$, inside the MOTS (its overrun grey to the death cross in (c)–(e)); level 6 from leg 1’s $t = 35$ checkpoint through the failure window; level 4 to $t = 100$ with no NaN. In-code Ψ_4 is identical on both overlaps. (a) Areal radius and (b) Misner–Sharp mass of the common MOTS. Gold: the spectral finder’s history (Sec. IIID, one row per plotfile, the three replay legs joined), born at $t = 18$ with $R = 5.634$ ($M_{\text{MS}} = 2.817$) and settling—monotone to 0.08% per row—onto the Schwarzschild sphere of the pair’s mass: $R = 4.7787$ and $M_{\text{MS}} = 2.3894$ at $t = 100$, 1.4% above $2M_{\text{ADM}}$ and 1.4% above M_{ADM} . Grey dashes: the round star scan, the inner bound of Sec. IIID, late and small at birth and blind over $t = 26$ – 35 (Sec. VIA). Rules: one isolated throat, $R_* = 3.8895$; $\sqrt{2}R_*$, a sphere with both throats’ area; $2M_{\text{ADM}}$ (M_{ADM} in (b)). (c) Separation of the χ pits. (d) $\max|K|$. (e) Grid maxima of $|\phi|$ (solid) and $|\text{II}|$ (dashed): the field is swallowed. Gold dotted rule: horizon birth. Movie: Video 3; youtu.be/3sx9GHRTYVs.

a level-6 leg walks through the failure window, and a level-4 leg carries the remnant to $t = 100$ with no NaN. The finder’s history is continuous across both seams (R steps by 0.08% and 0.04%).

The remnant is a black hole with both throats inside it, and it shrinks as it swallows the phantom field that held them open (Fig. 5(e)): R falls 15.2% from birth— 28% of the birth area gone—to $R = 4.7787$ at $t = 100$, rounding from deformation 0.104 to 0.004 , while M_{MS} falls from 2.817 to 2.3894 . Gravitational waves carry only 1.7% of that loss (the measured $E_{\text{GW}} = 1.2 \times 10^{-2} M$, Sec. VIIB); the rest is the negative-energy field going down the hole. The shrink is monotone on the finder’s 84 rows—no row rises on its predecessor by more than 0.08% —as the first law of Sec. IVE demands of a phantom influx, $T_{ab}k^ak^b < 0$, and that law is measured on this surface. Over $t = 51$ – 60 , replayed with every plotfile kept, the observed $\Delta R = -4.1 \times 10^{-3}$ is bracketed by the law evaluated on the found surface: -7.0×10^{-3} from the phantom influx alone, -2.5×10^{-3} with the outgoing shear’s growth term added. By $t = 90$ – 100 the influx is spent: the phantom rate on the surface sits 69-fold below its $t = 45$ – 55 value, the total first-law rate 112-fold, and

the radius drifts at 3.2×10^{-4} per unit. The late decline is exhaustion, not accretion.

The end state is the Schwarzschild sphere of the pair’s Bondi mass. At $t = 100$ the horizon sits 1.4% above $2M_{\text{ADM}}$ in radius and 1.4% above M_{ADM} in mass: with $R = 2(M_{\text{ADM}} - E_{\text{GW}} - E_\phi)$ and the measured E_{GW} , the offset puts $E_\phi = -0.039$, negative as the ghost channel radiates and of the sign and order of the flux integrals of Sec. VIIG. The $d = 6$ orbital merger’s remnant settles within 2.2% of the same radius (Sec. VII C), as it should if the end state is set by the pair’s mass.

VII. THE ORBITAL CHANNEL

A. Merger or fly-by

Varying the tangential momentum p at $d = 12$ on boosted, solved data separates scattering from plunge (Fig. 12 in Appendix B). The two slower arms scatter: at $p = 0.25$ the pair passes at 2.32 ($t \approx 48$) and recedes to 8.5 by $t = 100$; at $p = 0.45$ it passes closer, 2.18 at $t \approx 48$, and recedes to 5.3 . Both reach their stop

times with no NaN—the $p = 0.45$ arm at stock settings to $t = 100$, so the channel has no wall of its own below the plunge boundary—and the finder sees no trapped surface on any plotfile of either. The two faster arms plunge: $p = 0.60$ reaches contact at $t \approx 40$ (pit separation 1.86 at its first leg’s stop) and $p = 0.90$ just past its $t = 40$ leg (2.4 and still falling), each collapsing to a merged core on which no common MOTS converges (Sec. VIID). The plunge boundary therefore sits between $p = 0.45$ and 0.60 : 90–120% of the circular momentum under the combined pull, bracketing the circular value itself. That pull, $1 + Q$ times gravity’s (Eq. (5)), is central: a pair that misses contact keeps its angular momentum and swings past. But the approach deepens with speed—closest approach falls from 2.32 through 2.18 to contact, against the Newtonian expectation of a pericentre growing with $J = pd$ —so capture here is contact, not an innermost orbit. Nor do the scatterers escape: both start below the circular momentum and remain bound in Newtonian terms, and what follows a pass is set by the throats, which inflate (Sec. VIIB).

The recession after the turn is not that of two compact throats: on both scatterers the mouths inflate, and the constraint norm crosses the campaign’s 2.5×10^{-2} criterion as they do, at $t = 63.3$ ($p = 0.25$) and 67.6 ($p = 0.45$)—the trust windows past which nothing is quoted, from the runs or their waves (Sec. VIID).

B. The race between contact and inflation

At $d = 12$ no arm forms a common horizon at any momentum. Below the plunge boundary the pairs scatter (Sec. VIIA); the slowest tangential arm, $p = 0.12$, and an inward-momentum scout ($p = 0.25$ aimed straight at the companion) end the same way: the mouths inflate faster than the orbit closes—the scout’s $R_{\min}$ grows +4.7% by $t = 8$ and +72% by $t = 24$ while its infall stalls near separation 2.2—and the finder holds no common MOTS on any plotfile of either (the $p = 0.12$ arm dies of the interior failure at $t = 71.78$ having never trapped within its search window).

The mechanism is the branch selection of Sec. IVC, run by the companion. A lone throat’s fate is opposite to a declared kick: pushed outward it collapses, squeezed it inflates. A companion squeezes: each mouth starts in the isolated throat’s shape (Table I), yet grows as a lone throat kicked inward on the seed ladder of Sec. IVC, at $\varepsilon_{\text{eff}} \approx -0.4\%$ for $d = 12$ and -1.2% for $d = 8$, so every pair’s mouths start on the inflation branch, at any momentum, and grow on the throat’s own e-fold time $\tau \approx 5.88$ (Sec. IVB). That the solve keeps a declared seed is checked directly: the $\varepsilon = +0.1$ kicked single, run through the constraint solve, is born trapped (a MOTS on its $t = 0$ slice, $R = 4.306$), the fate its kick dictates. A merger is therefore a race against the inflation it starts: the head-on’s horizon arrives 3.1 e-folds in and wins; the $d = 6$ orbit’s arrives at 2.2 e-folds and wins (Sec. VIIC);

the $d = 12$ approach, 8 e-folds to its closest pass, always loses. The design rule for an orbital merger follows: short separation and small tangential momentum.

C. The $d = 6$ merger

The design point that wins the race is $d = 6$ with a small tangential twist, $p = 0.10$ per mouth. A level-3 scout merges, and the production chain repeats it on the $L = 128$ box (Fig. 6): the spectral finder holds a common MOTS around both mouths from $t = 13$ —2.2 e-folds into the inflation, the pits still 3.6 apart—born with $R = 5.60$ ($M_{\text{MS}} = 2.80$, deformation 0.115) and shrinking monotonically ($\leq 0.05\%$ per row over the finder’s 88 rows) by 12.8% to $R = 4.883$, $M_{\text{MS}} = 2.441$, deformation 0.016 at $t = 100$: a settled black hole within 2.2% of the head-on remnant’s radius, reached through an orbit. As in the head-on, the horizon is born common—no surface ever closes about either mouth—and it swallows the pair mid-inflation.

The chain crosses a harder wall (Sec. VIID) in four legs: level 5 from $t = 0$ to $t = 26.26$; a $\sigma_{\text{KO}} = 1.0$ leg to $t = 30.5$; the χ -floor leg ($\min \chi = 10^{-4}$), which sails past every earlier death to $t = 36.6$; and a level-4 settle leg, the wall knobs reverted, to $t = 100$ with no NaN. The floor acts only inside the horizon, over $t = 30$ –36.6; everything quoted comes from the finder’s surface outside it.

D. Interior failure after horizon formation

Two failure modes stand between a merged core and its settled remnant, and each has a measured cure. The first is a single-cell blow-up of the metric component h_{11} at the merged core’s centre: the head-on’s level-5 leg at $t = 38.845$, the $d = 6$ chain’s restarts at $t = 27.5$ –30.5, the $p = 0.60$ plunge at $t = 55.5$ —in every case with the global constraint norms clean to the last step. It is not under-resolution: on the $d = 6$ chain the level-6 restart dies at $t = 29.4$ and the level-7 restart dies *earlier*, at $t = 27.5$ with $\max |K|$ still climbing—a finer grid tracks the core’s collapse further and meets the failure sooner—and raising the dissipation to $\sigma_{\text{KO}} = 1.0$ delays the cell by a unit without curing it. What cures it is the χ floor: with $\min \chi$ capped at 10^{-4} the $1/\chi$ steepness runaway cannot complete, and both the $d = 6$ chain and the $p = 0.60$ plunge cross their death points and run to $t = 100$, the $d = 6$ settle leg then running floor-free. The second mode is the K wall: $\max |K|$ itself running away at the merging core, $93 \rightarrow 172$ in one step on the $p = 0.90$ arm at $t = 45.3$ with χ nowhere near its floor ($\chi_{\min} = 5.4 \times 10^{-3}$)—the floor does not touch it. Its knob is resolution: the $d = 6$ chain’s level-6 restart cleared the same runaway that stopped its level-5 leg.

Both failures are censored. The core fails only after a common MOTS encloses it: the $d = 6$ cell NaNs arrive

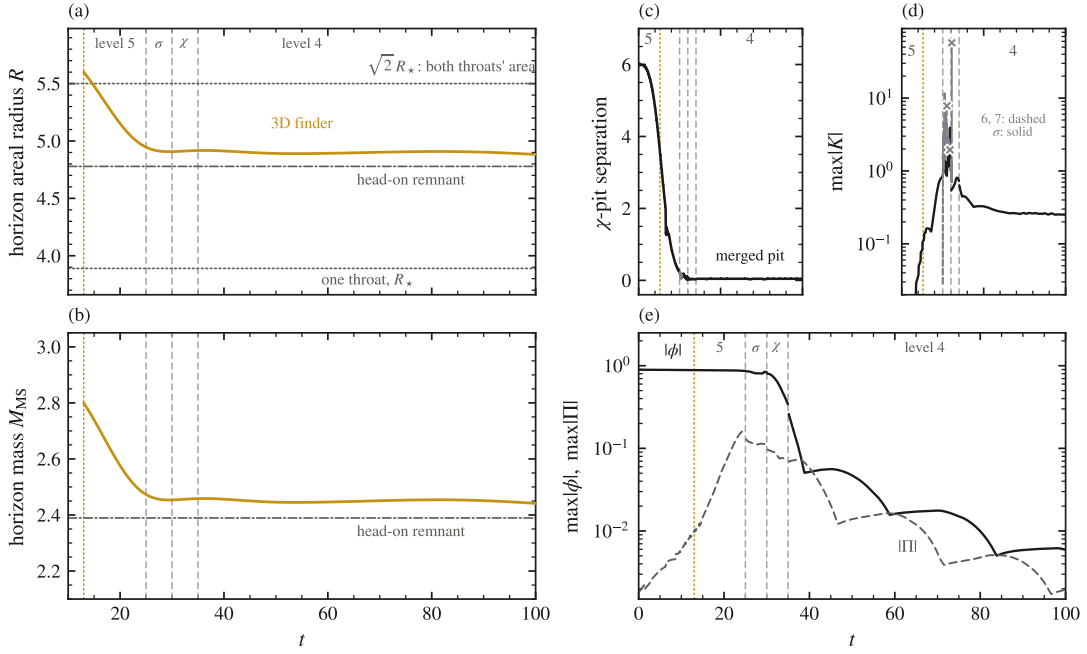


FIG. 6. The orbital merger ($d = 6$, tangential $p = 0.10$, opposite signs, boosted mode-3 data) carried through its merger wall, and the black hole it settles to. One evolution composed across its legs (faint dashed rules at the restarts, $t = 25, 30$ and 35 ; levels named in the panels, σ and χ for the two wall legs): level 5 from $t = 0$, stopped at $t = 26.26$ as the merged core's K ran away; the $\sigma_{\text{KO}} = 1.0$ leg; the χ -floor crossing leg; the level-4 settle leg to $t = 100$, no NaN. (a) Areal radius and (b) Misner–Sharp mass of the common MOTs: gold, the spectral finder's history (Sec. III D), born at $t = 13$ and settling from above onto the head-on remnant's values (dash-dotted rules, $R = 4.7787$; $M_{\text{MS}} = 2.3894$ in (b)). Dotted rules in (a): one isolated throat, R_* ; $\sqrt{2}R_*$, both throats' area. (c) Separation of the χ pits (grey: leg 1's overrun past its restart checkpoint). (d) The wall: $\max|K|$ of every arm—the used legs in ink; the level-6 and level-7 diagnostic restarts (grey dashed) die at $t = 29.4$ and, *earlier*, 27.5 , each a single-cell h_{11} NaN at the core's centre with the global norms clean to the last step; the σ arm (grey solid) clears the K wall and dies at the same cell at $t = 30.5$ (death crosses); the χ -floor and settle legs carry the history through. (e) Grid maxima of $|\phi|$ (solid) and $|\Pi|$ (dashed): the field is swallowed. Gold dotted rule: horizon birth. Movie: Video 4; youtu.be/jNnMAXPjyRA.

15–18 units after its horizon, the head-on's 21 after its, and lone collapses die behind their horizons at level 3 too ($\varepsilon = +10^{-3}$ at $t = 40.1$, 15 units after its horizon forms; $\varepsilon = +10^{-2}$ dressed with $\varepsilon_2 = 10^{-2}$ at $t = 27.6$, 14 after), while no level-4 lone collapse dies. The failure occupies one cell (the K wall, a few), the exterior is already set, and the finder's surface rides through every crossing unmoved: whatever diverges is the coordinate representation of a merged interior, behind the horizon.

The record thus tests weak cosmic censorship outside the physically reasonable matter it is posed for [95, 96]: in every collapse resolved here—the head-on across its legs, the $d = 6$ chain at levels 5–7, and the lone collapses—the evolution fails, if at all, only inside a MOTs. This is evidence for censorship under null-energy violation, not a proof—the failure is a coordinate blow-up read on the slice, not a located curvature singularity—and the plunges whose horizon never converges (Sec. VII A) leave their failures uncensored on the slice, at the edge of what the grid resolves.

VIII. GRAVITATIONAL WAVES

Every channel radiates, and at our extraction radii the most energetic is the fly-by at the capture boundary, which forms no horizon. The peak $|r\Psi_4|$ of the dominant mode at the innermost sphere is 6.1×10^{-2} for the fly-by, 5.1×10^{-2} for the plunge, 4.8×10^{-2} for the head-on, 2.1×10^{-2} for the $d = 6$ merger, 1.1×10^{-2} for the vacuum reference and 2.9×10^{-3} for the collapsing throat (Fig. 7). Each record propagates outward at $v/c = 0.86$ – 1.00 between neighbouring spheres, the correlated-lag measurement of Sec. VIII E (Fig. 7, right). No sphere is asymptotic: the dominant periods are comparable to the extraction radii, and the peak $r\Psi_4$ falls across the spheres by 22% on the head-on and 20% on the fly-by. We therefore do not extrapolate, and carry the spread over spheres as the near-zone error bar of every energy (Fig. 8(d)).

Every quoted burst arrives with its physical trigger—collapse, contact or closest pass, tens of units after the first light-crossing $t \approx R$ —and the per-sphere envelopes rise with it (Fig. 7, right). The solved data's own transient is small: on every binary the first light-crossing win-

dow, $t - R \leq 10$, carries at most 2.6% of the quoted band energy. On the seeded singles the wave is the throat's, not the seed shell's (Sec. VIII A).

Each channel is set against black holes at its own d and p (grey in Fig. 7(b)–(d)). The vacuum merger, at $d = 6$ and $p = 0.10$, peaks at the drainhole merger's retarded time, $3.0\times$ lower. The fly-by's control, at $d = 12$ and $p = 0.45$, is unbound in vacuum (circular 0.204, parabolic 0.289): the black holes start at periapsis and coast apart, radiating $1.05 \times 10^{-3} M$ in a single cycle (peak $|r\Psi_4| = 9.0 \times 10^{-3}$, $6.7\times$ below the fly-by's), whereas the drainhole pair falls from 12 to 2.18 under a pull six times gravity's and outradiates it by $40\text{--}70\times$. For scale, the closest non-merging black-hole scatterings computed in vacuum radiate $\approx 2 \times 10^{-2}$ of their total mass in all multipoles [78].

A. Collapsing throat

A lone throat radiates only if its collapse is not spherical. The spherically kicked arm emits nothing above the $\ell = 2$ numerical floor, while the same kick with an $\varepsilon_2 = 0.05$ quadrupole produces a burst of peak $|r\Psi_4| = 2.9 \times 10^{-3}$, $44\times$ the floor of the matched level-4 control at $R = 14$, carrying $2.6 \times 10^{-5} M$ (integrated to $t = 58$, before the numerical floor dominates). The burst arrives in radius order at $0.8\text{--}1.0c$, carries the same $r\Psi_4$ to four spheres within 7.2%, and is linear in the seed (Fig. 11 in Appendix A): $4.85\times$ for a $5\times$ kick, ratios $1.99/1.97/1.87$ at $R = 10/14/18$ for a halved seed, linear to 1–16% across $\varepsilon_2 = 0.005\text{--}0.05$. The pure quadrupole— $\varepsilon_2 = 0.01$ with the radial kick removed—radiates the same amplitude to 2%, its waveform leading the kicked arm's by $1.1\text{--}1.5 M$: the deformation sets the wave, the kick only its clock. The ringdown is read at $\varepsilon_2 = 0.05$: the 0.01 burst, matched to the radial kick, stands only $\approx 2\times$ above the spherical control's floor (Fig. 11(b)). The remnant rings at period $20.6 M$ (dashed in Fig. 7(a); an independent least-squares fit gives 20.61 ± 0.15), with no drift across the window (half-periods 9.4, 10.3 and $9.0 M$), consistent with the Schwarzschild fundamental of a black hole of mass 1.16–1.21 (period $19.6\text{--}20.4 M$ [80]), the horizon mass near its minimum, and not with the larger mass the horizon carries at the window's start (1.47, period $24.7 M$).

B. Head-on ringdown

The head-on's signal is a bell with no chirp: the (2,0) breathing mode at $R = 10$ swings through 4.8×10^{-2} , -4.1×10^{-2} , 2.5×10^{-2} at $t = 29, 44, 63$ —the half-period stretching from 15.5 to 18.8, the amplitude falling $\times 0.61\text{--}0.86$ per half-swing—and arrives ~ 4.4 units later per 4 units of extraction radius ($0.92c$), outgoing across all three spheres. On the detector track it is the bell that slows from 707 to 410 Hz in Fig. 8(c). Band-integrated

about its peak, the bell carries $E_{\text{GW}}/M = 1.2 \times 10^{-2}$ through $R = 14$, falling from 1.3×10^{-2} to 1.1×10^{-2} across $R = 10\text{--}18$, the bell's own near-zone spread. The vacuum control—two equal black holes released from rest at the same $d = 8$ —rings the same (2,0) bell at the same sphere $21\times$ quieter ($E_{\text{GW}}/M = 5.6 \times 10^{-4}$, against the published 5.5×10^{-4} for infall from rest at infinity [87]) and 28 units later (its crest at $t = 61$ against 33): without the scalar's pull the holes fall slower and hit softer.

C. Merger burst

The $d = 6$ merger's (2,2) burst is launched at contact—the common MOTS of $t = 13$ (Sec. VII C)—peaks at $|r\Psi_4| = 2.1 \times 10^{-2}$ on $R = 20$ at $t \approx 39$, folds onto one curve across $R = 20\text{--}44$, and the remnant's ringdown carries the record, clean, to $t = 100$ on the settle leg (Fig. 7(c)).

D. Scatter and plunge bursts

Each orbital arm emits one (2,2) burst at its closest pass or plunge, with no ringdown on the scatterers—no remnant forms—and the energy through the innermost sphere ($R = 20$) is not monotone in the momentum: $E_{\text{GW}}/M = 3.75 \times 10^{-2}$ at $p = 0.25$, 8.3×10^{-2} at $p = 0.45$, and 4.2×10^{-2} at $p = 0.60$, each in its own pair's solved mass (Sec. II D). The pair radiates hardest at the capture boundary (Sec. VII A), where the pass is deepest and slowest: the $p = 0.45$ arch carries $2.2\times$ the shallower scatterer's energy and $2.0\times$ the plunge's. Each scatterer's arch is integrated to the trough where the decaying burst meets the mouths' growing contamination ($t = 84$ and 94 at $R = 20$); the next sphere out reads 3.95×10^{-2} and 8.2×10^{-2} , the near-zone spread. The plunge's burst is cut by its run's trust window, 12 units past its $R = 20$ peak, so its energy is a floor and the boundary-to-plunge ratio a ceiling.

E. Vacuum twin

The chirping reference—a vacuum binary at $d = 12$, $p = 0.12$, equal masses, no scalar—behaves as a binary black hole must: it merges, climbs the point-mass chirp $299 \rightarrow 473$ Hz and turns toward the equal-mass Kerr mode (Sec. IX A). It also calibrates the wavefront method: its flat-topped merger envelope defeats peak timing, which reads $0.57c$ on it against the $0.82c$ of its whole-waveform lag, so every speed here is a correlated lag.

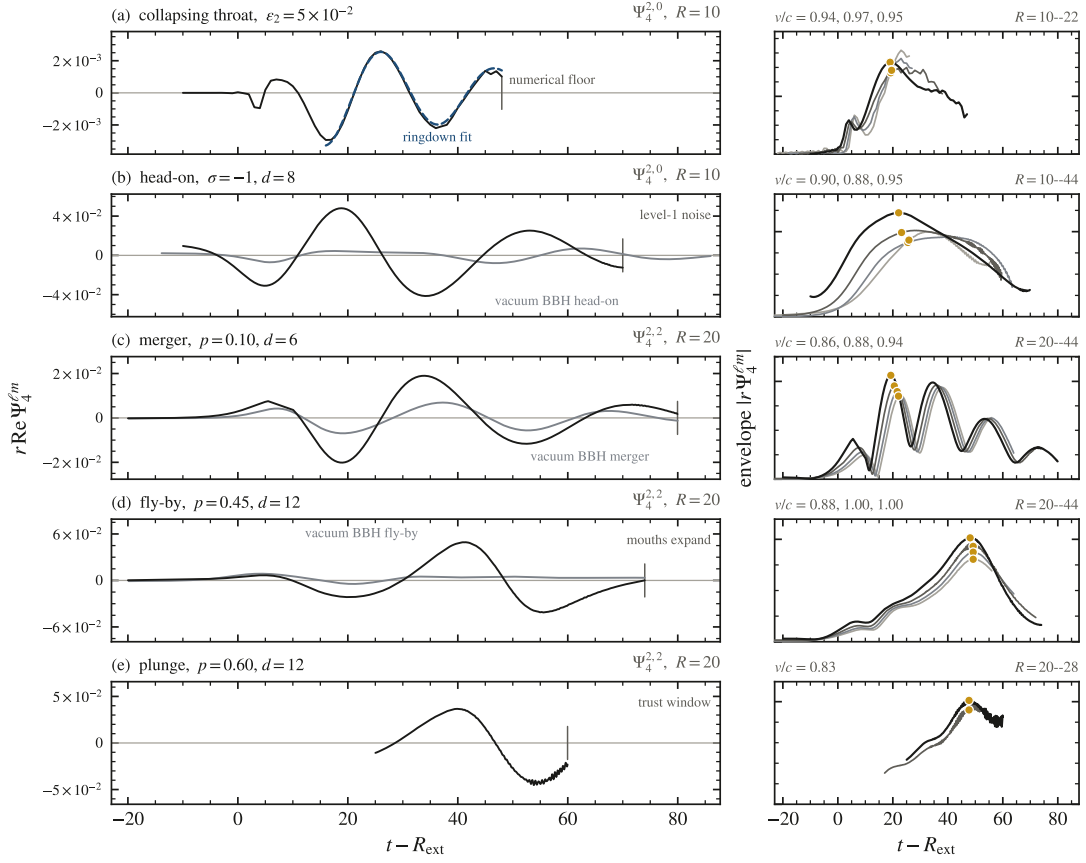


FIG. 7. The waveform gallery: one row per scenario on one retarded clock. Left: $r \text{Re } \Psi_4$ of the dominant mode at the innermost sphere, each row on its own scale; row (a) is the $\varepsilon_2 = 0.05$ arm, with the ringdown fit of Sec. VIII A dashed (period $20.6 M$). Caps mark where the drawn record ends: the collapsing throat where its numerical floor passes 10% of the burst peak; the head-on where the noise of its level-1 refinement cube reaches the sphere (from $t = 80$ at $R \leq 20$); the fly-by at the trough where the mouths' expansion meets the decaying arch ($t - R = 74$); the plunge at its trust window. The merger runs whole, burst through remnant ringdown. Grey under (b)–(d): the vacuum controls, black holes at the same d and p on bare punctures, on the same scale: the vacuum head-on rings $6.1 \times$ lower and crests later (Sec. VIII B); the vacuum merger peaks at the drainhole merger's retarded time, $3.0 \times$ lower; the vacuum fly-by radiates one cycle as the holes swing away from periapsis, $6.7 \times$ below the fly-by's arch. Right: the amplitude envelope at every sphere on its own retarded time (ink to faint, inner to outer)— $|r \Psi_4|$ for the complex $(2, 2)$ modes, the analytic-signal envelope for the real $(2, 0)$ modes—which folds onto one curve for a signal moving at c and falling as $1/r$. Gold dots: the correlated wavefront. The quoted v/c is the whole-waveform lag between neighbouring spheres (Sec. VIII E), the plunge's the burst-peak lag: throat 0.94/0.97/0.95, head-on 0.94/0.88/0.95, merger 0.86/0.88/0.94, fly-by 0.88/1.00/1.00, plunge 0.83. Movies of the vacuum head-on and fly-by: Videos 5 youtu.be/aRWvD4W8OOg and 6 youtu.be/65_3kw4obvE.

F. Scalar channel and its sign

Ψ_4 is blind to a spin-0 wave. Two opposite scalar charges—every merging pair here (Sec. V)—carry a dipole, where a vacuum binary's leading moment is the quadrupole. On every binary the scalar stream rides the offline Ψ_4 spheres (Sec. III C), so the two channels compare sphere by sphere (Fig. 13 in Appendix B). The recorded kinematic flux F_{kin} approximates the scalar energy crossing S_R per coordinate time: the flux, through the world tube of S_R , of the canonical energy current $J^\mu \equiv -T^{(\phi)\mu}_\nu t^\nu$ with $t^\mu = \alpha n^\mu + \beta^\mu$, conserved where t^μ is Killing and nearly so in the quasi-stationary exterior

the spheres sample [52, 53],

$$\begin{aligned} F_\phi^{\text{geo}} &= \oint_{S_R} \alpha s_i J^i \sqrt{\sigma} d^2x \\ &= - \oint_{S_R} \partial_t \phi (\alpha s^j \partial_j \phi + \beta_j s^j \Pi) \sqrt{\sigma} d^2x, \end{aligned} \quad (13)$$

positive for an outgoing canonical wave (s^i the sphere's outward unit normal, $\sqrt{\sigma}$ its area element, $\partial_t \phi = \alpha \Pi + \beta^i \partial_i \phi$). On a conformally flat sphere with uniform α and χ and no shift, $F_\phi^{\text{geo}} = \alpha^2 \chi^{-1/2} F_{\text{kin}}$; at $R = 30$ the initial data's area mean of $\alpha^2 \chi^{-1/2}$ is ≈ 0.94 , so energies built from F_{kin} overstate $|F_\phi^{\text{geo}}|$ by about 6%; the shift terms are neglected.

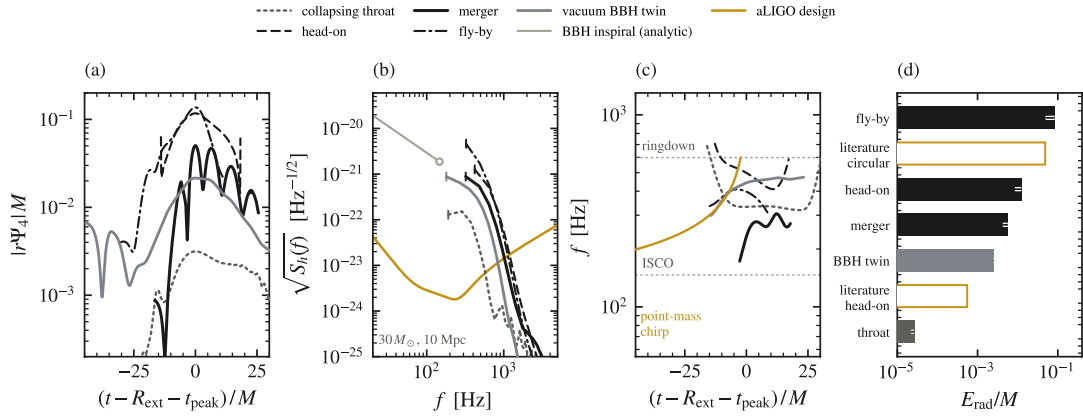


FIG. 8. Every source at one calibration: total mass $30 M_\odot$ at 10 Mpc, each record in units of its own total ADM mass. (a) Envelopes $|r\Psi_4|/M$ at the innermost sphere, aligned at each peak: fly-by loudest (1.4×10^{-1}) to lone throat (3.1×10^{-3}). (b) Strain amplitude spectral densities over the aLIGO design floor (gold), each from its own low-frequency corner (one cycle per record, ticked). The curves are falling power laws (slopes -3.4 to -5.8) with no resolved maximum, so any “peak” read here is the corner $1/T$. Faint line: the Newtonian inspiral of the same binary, $|\dot{h}| \propto f^{-7/6}$ to its ISCO at 147 Hz. (c) Instantaneous frequency against the point-mass chirp (gold), the ISCO (147 Hz) and the equal-mass Kerr ringdown (599 Hz): the vacuum twin climbs the chirp and peaks at $fM = 0.065$; no wormhole channel does (Sec. IX A). (d) Energy radiated in the dominant multipole; ticks give the spread over spheres (fly-by 41%, the widest). Open gold bars: published equal-mass vacuum values: head-on infall from rest at infinity, 5.5×10^{-4} [87]; quasi-circular inspiral from infinite separation and merger, 4.84×10^{-2} [81]; all multipoles. Energies are band integrals of $|\int r\Psi_4 dt'|^2$ over a common retarded window (the lone throat’s closing at $t = 58$, before its floor), doubled for $\pm m$; gravitational channel only. At 10^4 – $10^6 M_\odot$ and $z_e \simeq 20$ the same records are millihertz LISA sources (Sec. XB, Fig. 9). Movie of the vacuum twin: Video 7; youtu.be/NkcpGiJlxXI.

The channel is dipole-dominated and comparable to the gravitational one. On the fly-by at $R = 30$, $\ell = 1$ carries the scalar power $\sum_{\ell m} |\dot{A}_{\ell m}|^2$ to one part in $10^{6.2}$, and the dipole leaves as a single arch that crests at $t = 47$ on $R = 14$ before $t = 60$ on $R = 30$ —an outward pulse, and one this run holds complete: the outer sphere’s trusted record ($t - R \leq 67.6$) ends only after the arch has passed (Fig. 13(a)). Over that whole record $|E_\phi| = 4.7 \times 10^{-2} M$, 0.56 times the gravitational energy that crosses the sphere with it, and the estimator stops mattering: the band-limited energy of Fig. 8(d) gives 0.61 and the wave-zone mode sum 0.60. Part of that energy is not radiation: before the arch F_{kin} rises on every sphere at once, ahead of any signal from the throats, and falls with radius as $R^{-2.5}$ —the near-zone field of the moving charges, not a propagating wave. It supplies 14% of $|E_\phi|$, and the arch alone gives $|E_\phi|/E_{\text{GW}} = 0.48$. Mid-burst, at $t = 60$, the same three read 2.4, 3.7 and 3.1, and the inner sphere 0.6 (Fig. 13(c)), so downstream we use only the channel’s multipole, direction, sign and order of magnitude.

Gravity couples to minus the canonical stress tensor, Eq. (1), so the physical energy flux is $-F_{\text{kin}}$: the outgoing $\ell = 1$ wave—on the fly-by and, once their horizons form, on the head-on and the orbital merger—carries negative energy. If the ADM energy balance holds with this term, a bound pair emitting it gains energy and radiation reaction is *anti-damped*: the classical face of the standard objection to ghosts, whose quantum vacuum decays into ghosts and positive-energy quanta together at a rate kept finite only by cutting the effective theory off [71, 97].

Classically these binaries make the same trade, and the head-on remnant settles near the pair’s Bondi mass, a little above $2M_{\text{ADM}}$ (Sec. VIB).

The dipole returns angular momentum too: on the fly-by, through both spheres and over its trusted record, the scalar’s physical angular-momentum flux is opposite to the gravitational waves’, which carry off angular momentum of the orbit’s sign. Nor does the negative energy invert the nonlinear memory [101–103]: per unit energy an equatorial dipole projects onto the $(2, 0)$ memory with the opposite sign to a quadrupole, so the ghost’s negative flux adds to the gravitational memory, by 0.30–0.46 of it on the fly-by at $R = 30$ across the scalar estimators.

G. Horizon censorship of the scalar channel

The head-on’s chain carries the scalar stream end to end, and its horizon switches the channel off (Fig. 13 in Appendix B). After the common MOTS closes ($t = 18$), the mouths’ remaining hair leaves as a single $\ell = 1$, $m = 0$ pulse, cresting later at each sphere in causal order ($t = 34$ and 39 at $R = 14$ and 18), and the envelope then decays: log-linear fits over $t = 30$ – 95 give $\tau = 20/25/32$ at $R = 10/14/18$. The decay is the remnant ringing down its hair: fitted from $t = 40$ as one damped oscillation, the dipole at $R = 10$ rings at $\omega = 0.123$ and decays at 0.037, against 0.1226 and 0.0409 for the $\ell = 1$ scalar quasinormal mode of a Schwarzschild hole of the final MOTS mass ($M\omega = 0.2929 - 0.0977i$ [80]): the

frequency agrees to 1% for fits started anywhere over $t = 35$ –45, and the damping is 9% slower. The post-horizon energies, integrated from the MOTS, keep the sign of Sec. VIII F, $E_\phi = -0.0294/-0.0525/-0.0600 M$. Only that window is integrated: before $t = 26$ the flux at $R = 10$ runs canonically ingoing, the two open mouths’ static hair overlapping and draining inward.

The orbital merger, the one other encounter with a horizon (common MOTS from $t = 13$; Sec. VII), sheds the same way on its own glued record: its dipole crests at $t = 39$ on $R = 20$ and falls exponentially, $\tau = 22$ (fit over $t = 45$ –95), $19\times$ down at $R = 14$ by the record’s end. Without a horizon there is nothing to switch the channel off: the fly-by radiates its arch as the mouths swing past one another, and what ends that emission is the encounter itself—the mouths leave intact, still carrying the charges that source the dipole.

The lone throat shows the same contrast. Re-run at level 4 with the scalar stream, the $\varepsilon = +10^{-2}$ throat collapses on schedule—a permanent MOTS from $t = 11$, R settling $3.88 \rightarrow 2.54$ —and its scalar envelope, after the burst, falls by 30 – $100\times$ across the spheres by $t = 80$, with late e-fold times of 6 – 7 . The unkicked level-4 throat, which inflates with no trapped surface by scan or flow finder, instead grows quasi-exponentially in the $\ell = 0$ monopole (e-fold ≈ 5 , shortening across $t = 45$ –75), identically in a box-doubled ($L = 128$) copy through $t \approx 80$, so the growth is the throat’s and not the boundary’s or the sponge’s; beyond $t \approx 80$ the constraint norms grow, and only $t \lesssim 80$ is counted. The contrast is confounded with the fate: the horizonless arms are also those whose mouths inflate toward the spheres. What is measured is that the scalar channel shuts off behind a horizon and grows while a throat inflates.

IX. PATTERN MATCHING AGAINST DETECTOR DATA

In 2.25 h of coincident Advanced LIGO O3b data [30], a matched-filter search [27, 29] with 154 templates—the waveforms of Sec. VIII rescaled as $h \propto M/D$, $f \propto 1/M$ —finds *no candidate at zero lag*, and none was expected. No static throat of astrophysical mass survives (Sec. IV G), so LIGO could catch only the burst of an encounter or collapse at formation, never a wormhole in residence; and a population formed at $z_e \gtrsim 20$ would convert there, far beyond the search’s reach (Sec. X B). At single-detector SNR 8 and optimal orientation the median horizon over each mass ladder is 5.1 Gpc (fly-by), 3.0 (head-on), 2.7 (merger), 1.7 (plunge, to its trust window), 1.3 (vacuum twin) and 0.15 (collapsing throat); the first four are upper bounds, since their strain, from fixed-frequency integration [28], ends at 0.10–0.48 of peak. This short search demonstrates the pipeline; much stronger limits on short bursts follow from the LVK O3 all-sky burst search [82], whose rate-density limits against characteristic frequency apply directly to the energies of Sec. IX B.

The null result is not a blind spot. Cut to the signal’s duration, binary-black-hole templates (IMRPhenomD) match the bursts with fitting factors of 0.893–0.995, as they match the vacuum control (0.935–0.951), so burst morphology alone cannot tell the two apart. All 18 injections into O3b strain at optimal SNR 20, each midway between two bank rungs, are recovered at 92–123% of optimal within 11 ms, and the χ^2 test vetoes none; the highest is a plunge recovered by a heavier rung, whose lower corner also collects the injected strain below the plunge record’s own.

A. The frequency track

The frequency track separates a drainhole burst from a binary black hole by what comes before the ringdown (Fig. 8(c)). The vacuum twin peaks at $fM = 0.065$ and climbs the point-mass track before turning toward the equal-mass Kerr mode at $fM = 0.0885$ [79, 80]: a monotone rise, then a fixed ringdown frequency. No drainhole channel follows it. The head-on’s frequency *falls*, by $\times 0.58$, from contact onto the $\ell = 2$ quasinormal mode of a Schwarzschild hole of its final MOTS mass, 397 Hz at this calibration, which it reaches to 3% by the end of its loud record: the remnant does not spin, and the shrinking of its horizon would raise that mode, not lower it. The fly-by’s frequency falls as the pair separates; the merger’s rises after contact and stays below its remnant’s mode through its loud record; the lone collapse rings at its own Schwarzschild mode (Sec. VIII A), 0.55 of the equal-mass Kerr one. Wherever a remnant’s ringdown is resolved—the lone throat’s, the head-on’s and the head-on’s scalar dipole (Sec. VIII G)—it is the ordinary quasinormal spectrum of the black hole left behind, as in any catalogue event [31]; the wormhole is in the burst before it.

B. Energy per event

The detector-independent deliverable is the energy per event in throat-mass units, in the dominant multipole about each source’s axis (sphere spread in parentheses): fly-by $4.9 \times 10^{-2} M$ at the outermost sphere (8.3×10^{-2} at the innermost), merger $5.7 \times 10^{-3} M$ (25%), head-on $1.2 \times 10^{-2} M$ (31%), collapsing throat $2.6 \times 10^{-5} M$ (14%), vacuum twin $2.6 \times 10^{-3} M$ (5%). Since Ω_{GW} scales as event density times energy per event, these convert any bound on Ω_{GW} [25, 26] into a bound on the abundance of primordial drainhole encounters, provided the ghost scalar exists at that epoch. The scalar channel adds comparable energy of opposite sign (Sec. VIII F).

X. DISCUSSION

All four fates of Sec. I occur, each with a measured selection rule: the relative scalar sign separates scattering from merger (Sec. V), the seed’s sign separates collapse from inflation (Sec. IV), and the orbital momentum separates plunge from fly-by at 90–120% of circular (Sec. VII). For detection, three results combine: no inspiral exists (Sec. XA), the wormhole phase is a burst with no chirp ahead of it, its frequency falling or drifting until the remnant’s ordinary ringdown takes over (Sec. IXA), and the scalar counterpart decays once a horizon forms (Sec. VIIG). A surviving wormhole signature is a drifting-frequency burst with a scalar counterpart; a settled ringdown without one is already the descendant.

A. No wormhole inspiral

A drainhole binary cannot radiate an inspiral. Each throat’s unstable mode gives it a lifetime $t \approx \tau \ln(1/\varepsilon)$, with $\tau = 5.3\text{--}5.9$ in isolation. An inspiral needs many orbits. Under the combined pull of Eq. (5), six times gravity at large separation and softened as $(d + \delta)^{-2}$ at the separations evolved (Sec. VB), a circular orbit at $d = 12$ has period $P \approx 75\text{--}100$. No pair starts quiet enough to wait that long: the companion itself plants the seed—the effective inward kick of 0.4% at $d = 12$ (Sec. VIIB)—an ε worth $t \approx \tau \ln(1/\varepsilon) \approx 29\text{--}32$ units—under half a period. Even $\varepsilon = 10^{-15}$ buys only 1.8–2.7 periods, and each further factor of ten adds only $\tau \ln 10 \approx 12\text{--}14$ units. The orbit’s demand grows with separation as $P \propto d^{3/2}$, while the throat’s endurance grows only as $\ln(1/\varepsilon)$, and the companion’s kick is part of the binary itself.

Radiation reaction cannot close the gap either: the leading radiative moment, the scalar dipole, carries negative energy comparable to the gravitational channel’s (Sec. VIIF), which, if the ADM balance holds, pumps the orbit rather than draining it.

The wormhole-specific signal is therefore a short transient: within a few tens of M each mouth collapses or inflates, and the plunge, burst and conversion of Secs. VI–VIII are all that this source class emits *as wormholes*; any long chirp belongs to the objects the throats become. The argument holds while the throat’s unstable mode exists, for the minimally coupled ghost scalar without spin of the González-Guzmán-Sarbach analysis. Rotation can remove that mode—in five dimensions above a critical angular momentum [55], in four as perturbative analyses indicate [56–58], with spinning Ellis wormholes existing up to an extremal limit [54]—and other supports are radially stable [59, 60], so a spinning or differently supported pair that outlives an inspiral is not excluded here. The same lifetimes rule out a static ghost-supported throat as a long-lived black-hole mimicker of the kind ringdown and echo tests target [98, 99].

B. Heavy seeds and LISA bursts

The lifetimes of Table IV forbid any throat from surviving from the early universe, so what follows is conditional on the epoch z_e at which throats of mass M would form, which our evolutions do not fix. The known mechanisms—bubbles and domain walls [5, 6, 89, 90]—form them at horizon re-entry, $t \sim 1$ s for $10^5 M_\odot$ ($z \sim 10^9\text{--}10^{10}$), where these bursts would be redshifted out of every band; we know of none forming $10^4\text{--}10^6 M_\odot$ throats as late as $z_e \approx 20$.

If such throats existed, collapse—alone or in a merger—would seed black holes. Wormholes pinching off at the mouths of inflating bubbles or domain walls leave black holes that can seed supermassive ones [5, 6, 89, 90]; here the wormhole is traversable, and its conversion is driven by its own instability or by a merger, in seconds to minutes (Table IV), at a remnant mass of order its progenitors’: the head-on and orbital remnants end at $R = 4.7787$ and 4.883 from pairs of ADM mass 2.3573 and 2.371 . The scalar hair leaves on a $\sim 20 M$ clock (Sec. VIIG), and the remnants then grow as ordinary black holes: from $10^4\text{--}10^5 M_\odot$ at $z \simeq 20$, Eddington-limited growth (Salpeter e-fold $\simeq 45$ Myr) reaches $10^9 M_\odot$ in $414\text{--}520$ Myr, inside the ~ 600 Myr available by $z \simeq 7$ —a heavy seed [50] (Fig. 9(a)) that needs no atomic-cooling halo, Lyman–Werner flux or baryonic environment, and that, predating structure, would make UHZ1’s overmassive morphology generic [44, 45]. One seed per present-day massive galaxy is $n \simeq 10^{-4}\text{--}10^{-2} \text{ Mpc}^{-3}$ comoving, a dark-matter fraction $3 \times 10^{-11}\text{--}3 \times 10^{-7}$; the FIRAS μ -distortion bound [51] on curvature-perturbation black holes of $10^4\text{--}10^{13} M_\odot$ does not apply, but only because the formation mechanism is left unspecified.

The encounters would radiate in LISA’s band (Fig. 9(b)): with the measured $(fM)_{\text{peak}} = 0.04\text{--}0.07$, $f_{\text{obs}} = (fM)_{\text{peak}}/[M(1 + z_e)]$ spans $0.4\text{--}70$ mHz across $10^4\text{--}10^6 M_\odot$ at $z_e \simeq 20$ (luminosity distance ≈ 230 Gpc [66]). Against the analytic LISA noise [64], sky and inclination averaged and counting only each record’s resolved band, the fly-by and the $d = 6$ merger reach optimal signal-to-noise ratios of $103\text{--}394$ at $10^5\text{--}10^6 M_\odot$ and the head-on $119\text{--}159$; all three stay above 8 from 3×10^4 to $6 \times 10^6 M_\odot$. Heavier events are louder but redder: the fly-by and the head-on hold 8 to $8 \times 10^6 M_\odot$, and at $10^7\text{--}10^8 M_\odot$ only the plunge still reaches it, to at least $4 \times 10^7 M_\odot$ (its record is cut at its trust window). A lone collapse, radiating only $2.6 \times 10^{-5} M$, stays below detection, at a signal-to-noise ratio of at most ≈ 6 . The merger and head-on bursts each announce a seed; the fly-by, the loudest, announces none. What limits detection is occurrence: every seed converts once, so LISA records $1.9 \times 10^3 (n/\text{Mpc}^{-3})$ bursts in four years, $\sim 0.05\text{--}5$ per year for $n = 10^{-4}\text{--}10^{-2} \text{ Mpc}^{-3}$ —resolved bursts, never a stochastic background, although their time average, $\Omega_{\text{GW}} \sim 10^{-13}\text{--}10^{-12}$ at ~ 7 mHz, would lie above LISA’s power-law-integrated sensitivity [65] for $n \gtrsim (6\text{--}36) \times 10^{-4} \text{ Mpc}^{-3}$ (Fig. 9(c)). The spectrum

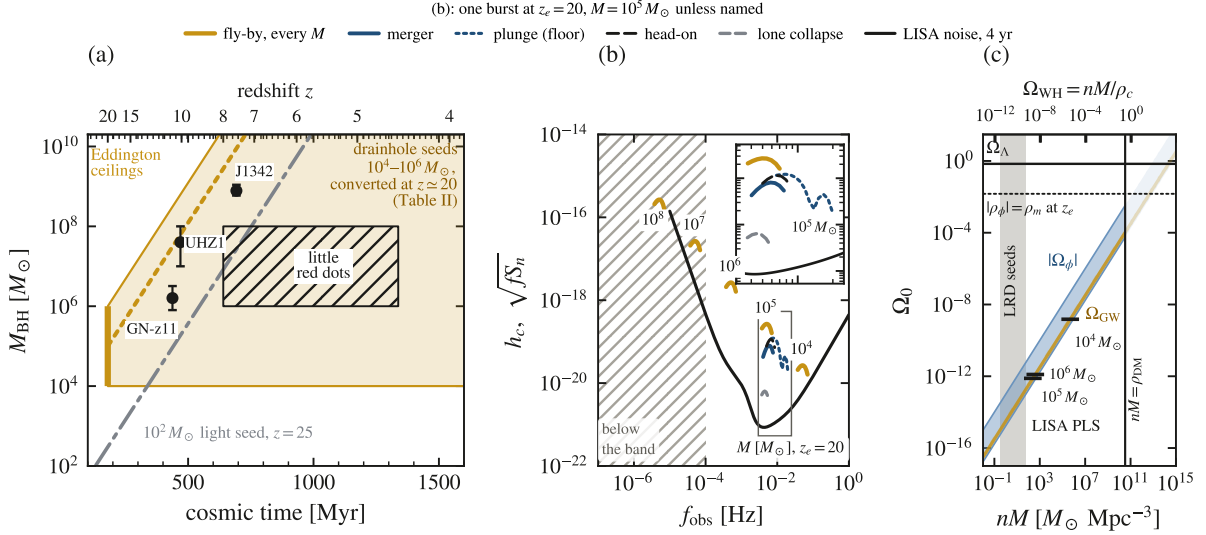


FIG. 9. The heavy-seed channel from measured numbers. (a) Seed growth: a drainhole population of 10^4 – $10^6 M_\odot$ converts at birth ($z \simeq 20$, seconds to minutes; Table IV), so its reachable region (gold) runs from the lightest seed idling to the heaviest seed's Eddington ceiling (Salpeter e-fold 45 Myr). UHZ1 [44, 45], GN-z11 [48], the $z = 7.5$ quasar J1342+0928 [49] and the little-red-dot box [46, 47] lie inside it; the first three lie *above* the ceiling of a $10^2 M_\odot$ light seed started at $z = 25$ (grey), which misses UHZ1 by two decades. (b) One burst against LISA: the characteristic strain $h_c = 2f|\dot{h}|$ of a fly-by (gold) at each mass decade from 10^4 to $10^8 M_\odot$ (exponents at the tracks), and of the $d = 6$ merger (blue), the $p = 0.60$ plunge (blue dotted; a floor, its record cut at its trust window), a head-on (dashed ink) and a lone collapse (grey) at $10^5 M_\odot$, magnified in the inset, all at $z_e = 20$, against $\sqrt{fS_n}$ of the analytic LISA noise [64] with the four-year galactic confusion (ink); inclination averaged, each track drawn over the band holding 90% of its $\int h_c^2 d \ln f$; the $10^7 M_\odot$ fly-by's counted signal lies in its tail above the band floor, past the drawn core. The squared ratio of track to noise, integrated over $\ln f$, is the burst's squared signal-to-noise ratio, so a short track can sit above the noise and still fall short of 8, as the lone collapse does. Heavier bursts are louder but redder: the tracks climb a decade per mass decade, the noise climbs faster below 0.1 mHz (hatched: below LISA's band, not counted), and the $10^8 M_\odot$ burst lies wholly below it. (c) Ω_{GW} of the conversions (gold, E/M from the head-on to the merger) and the negative scalar deposit $|\Omega_{\phi,0}|$ (blue, $|E_\phi| = 0.56$ – $3.7 E_{\text{GW}}$ with E_{GW} from the merger to the fly-by: every encounter, the case most favourable to Λ) today against the comoving mass density nM of converting wormholes, on which alone both depend (top axis: $\Omega_{\text{WH}} = nM/\rho_c$). Solid up to the dark-matter ceiling $nM = \rho_{\text{DM}}$ (vertical rule), faint beyond it: Ω_Λ is reached only at $\Omega_{\text{WH}} \simeq 60$ – 800 , above the cap $|\rho_\phi| = \rho_m$ at z_e that $H^2 > 0$ imposes (dashed). Ink ticks: the four-year power-law-integrated LISA sensitivity [65] at each mass's conversion frequency $(fM)_{\text{peak}}/[M(1+z_e)]$. Grey: one seed per little red dot [46, 47] at 10^4 – $10^6 M_\odot$.

is a pure burst population with no $f^{2/3}$ inspiral ramp (Sec. X A).

The negative-energy scalar radiation dilutes like radiation ($w = 1/3$) and has the wrong sign to act as dark energy; within the dark-matter budget its present density stays below 3×10^{-3} of critical. Nothing in the astrophysical mass range survives (Sec. IV G), so every observed supermassive black hole is at most a descendant of this channel, not a surviving wormhole mouth [91, 92]. Whether it is *the* channel behind the JWST population is a question of abundance, and the abundance is cheap: one conversion per little red dot, $n \simeq (3\text{--}6) \times 10^{-5} \text{ Mpc}^{-3}$ [46, 47], is a fraction 10^{-11} – 2×10^{-9} of the dark matter, and LISA would record 0.06–0.1 such bursts in four years: it tests the channel only if conversions outnumber the little red dots at least tenfold.

C. The inflating half of the population

Seeding is half the census. The branch is chosen by the perturbation's sign (Sec. IV C), so a sign-symmetric seed distribution converts about half of a population of lone throats and sets the other half inflating. For pairs born close enough to collide, let a fraction $f_<$ be born below the capture boundary, at 90–120% of the circular momentum (Sec. VII A). With the relative sign random, the like-signed half scatter and meet lone fates, while the opposite-signed half merge below the boundary and fly by above it, both mouths inflating. Per throat, $f_</math> then converts by merger, a quarter by lone collapse, and $3/4 - f_</math> inflates: between a quarter and three quarters of a paired population ends in black holes. An inflating throat creates, from regular data, an expanding anti-trapped region (Sec. IV F), which under the null energy condition would need an initial singularity [68, 69];$$

unlike the false-vacuum bubbles of those references and the domain-wall wormholes of Ref. [6], whose inflating interiors pinch off behind a horizon, it lies on our side of the throat and grows into the surrounding space, as the massless throat’s explosion does [13]. By $t = 218$ the neck accumulates 1.3 areal e-folds and the boundary 3.0, against the ~ 60 of cosmic inflation [70]: the record shows the mechanism, not the magnitude, and the seed channel of Sec. XB requires either that such regions stall or that the inflating half is rare.

D. Topological censorship

Topological censorship forbids an observer at infinity from probing the topology of a globally hyperbolic, asymptotically flat spacetime that satisfies the null energy condition [100]. A traversable wormhole evades it only by violating that condition, and the two branches end the evasion in opposite ways. On the collapse branch, alone or in a merger, the throats end inside a MOTS while the phantom outside is swallowed or radiated away (Secs. IVE, VIB and VIIG): the wormholes are not removed—no classical evolution changes the number of asymptotic ends [16]—but end hidden behind a horizon, as the theorem would have them. On the inflation branch no trapped surface forms and the throat stays open (Sec. IVF).

E. Scope

The results hold for one matter model and one family of data: a single minimally coupled ghost scalar, equal masses, one throat compactness ($m/a = 1/2$) and no spin, on constraint-solved, matched initial data whose momentum is the exact boost (Secs. IID and IIE). Horizons here are MOTSs on the evolved slices, which for matter violating the null energy condition need not lie inside event horizons (Sec. IIID). The cosmology of Sec. XB takes the birth abundance $n(M)$ and the conversion epoch z_e as inputs—Fig. 9(c) maps any choice onto its signals—and requires the ghost scalar to exist at that epoch.

XI. REPRODUCIBILITY AND COST

Every evolution in Table III is one line in a registry shipped with the results pack—its single knob against a named reference, its outcome, its caveats—and every figure is regenerated by a script in the pack from the archived diagnostic streams. The code, the analysis and the results pack are in the GRTeclyn [39] fork at <https://github.com/Nikchik-coder/GRTeclyn> (branch `feature/merger`): the drainhole initial data with its constraint solve and exact boost, the seeds of

Sec. IIF, the scalar damping and the in-code diagnostics in the evolution code; the spectral MOTS finder, the orientation-corrected surface scan, the figure scripts and the claims ledger that recomputes every number quoted here in the analysis layer; and the parameter files and thinned diagnostic streams of every run in the pack. Each run’s manifest names the binary it ran, and a table in the pack maps every binary to its source commit. The evolutions are bit-reproducible on the production hardware: two builds differing only in a passive diagnostic give byte-identical physics streams over 2384 output rows.

Every evolution ran on one NVIDIA H100 80 GB device, with GRTeclyn built CUDA-native against AMReX 26.02. At $\Delta t = 0.02 \Delta x$ a production arm ($L = 64$, 128^3 base, three refinement levels) advances 6–18 code units per hour, and a $t = 100$ single-throat or head-on evolution takes 5–7 h. Depth is the expense: level-5 arms run at 2–4 units per hour on the full box (the octant $L = 512$ arm, an eighth of the cells, at 28–74, 11.9 h in all)—29.3 h for the head-on’s three-leg $t = 0 \rightarrow 100$ chain, 46.3 h for the $p = 0.25$ fly-by at level 5, which peaks at 58 of the 80 GB, and 26.9 h for the $d = 6$ merger chain. The 96 evolutions total 476 GPU-hours, each run’s cost recomputed from its packed logs: the logged speed and time span, a restarted leg dated from its checkpoint.

ACKNOWLEDGMENTS

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Appendix A: Code Health and Constraint Evolution

Figure 10 gives the constraint record of every run the main-text figures draw, and Fig. 11 the linearity of the collapsing throat’s wave in its quadrupole seed (Sec. VIIA).

Appendix B: Supplementary Measurements

Three measurements support the main text without being needed to follow it: the momentum scan that separates plunge from scattering (Fig. 12; Sec. VIIA), the scalar channel’s energy, multipoles and validity window on the Ψ_4 spheres (Fig. 13; Sec. VIIF), and the horizon’s shut-off of that channel (Fig. 13; Sec. VIIG).

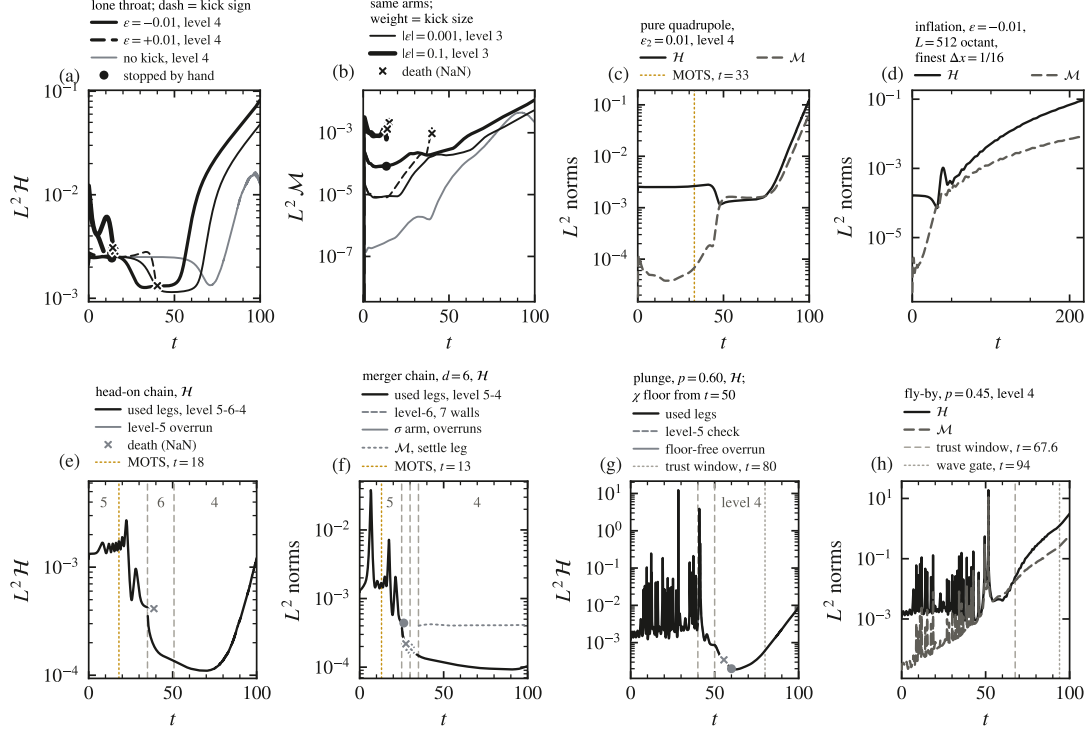


FIG. 10. Constraint evolution of every run the main-text figures draw, each at the highest refinement level it was run at (named in each panel's key): L^2 norms of the Hamiltonian ($\mathcal{H}$, ink) and momentum ($\mathcal{M}$, grey dashed) constraints, box averages on the base grid ($\Delta x = 0.5$; 2 on the octant of (d)) that compare only within one box (Sec. III C). (a), (b) The lone throat of Fig. 2: level 4 for the unkicked throat and $\varepsilon = \pm 10^{-2}$ ($+10^{-2}$ stopped by hand, dot), level 3 for $\pm 10^{-3}$ and ± 0.1 (dash = sign, weight = amplitude, grey = no kick, crosses = deaths). At $t = 0$ the norms carry the exact data's discretisation floor plus the kick's own defect, $0.090|\varepsilon|$ (Sec. II F), which feeds $\mathcal{M}$ within the first steps; the $|\varepsilon| \leq 10^{-2}$ arms then follow the unkicked norms to 5.4% until the fates part, while $\varepsilon = \pm 0.1$ starts 3–6 $\times$ high and dies in a one-step overflow at the compactified origin. (c) The pure-quadrupole collapse of Fig. 3 (gold, its MOTS from $t = 33$): the late rise is a grid mode that starts on the refinement boundaries, as on every long single-throat arm, and a doubled box does not delay it. (d) The inflating throat of Fig. 3 to $t = 218$. (e) The head-on chain of Fig. 5 with its ladder (era rules at the restarts, levels per era): the level-5 leg's overrun dies of the censored interior NaN at $t = 38.845$ (cross) with its norm below its value at formation (gold rule, $t = 18$). (f) The $d = 6$ merger chain of Fig. 6: the used legs in ink, every failed arm in grey to its death cross (the level-6 and level-7 restarts dashed; times in Fig. 6(d)). At every end, hand stop and death alike, the norms sit at or below their start: each failure is the single-cell interior NaN of Sec. VII D, not a constraint blow-up. (g) The $p = 0.60$ plunge, level 4 throughout: the floor-free extension (grey) dies of the merged core's cell NaN at $t = 55.5$ with its norm at its floor, the χ -floor leg carries the record to $t = 100$, and a level-5 restart (grey dashed) rides the merger clean; dotted, the trust window ($t = 80$). (h) The $p = 0.45$ scatterer: the norms sit on their floor through the pass and grow as the mouths inflate; rules, the trust window ($t = 67.6$) and the wave gate ($t = 94$). The needles on (g), (h) and the $p = 0.25$ arm, never on the head-on or $d = 6$ chains, are the base-grid norm reading the moving throats' χ pits: on the $p = 0.25$ verification pair four cells next to the pits carry 99.99% of a needle while the level-3 solution is clean, and neither $\max|K|$ nor the waves carry a counterpart.

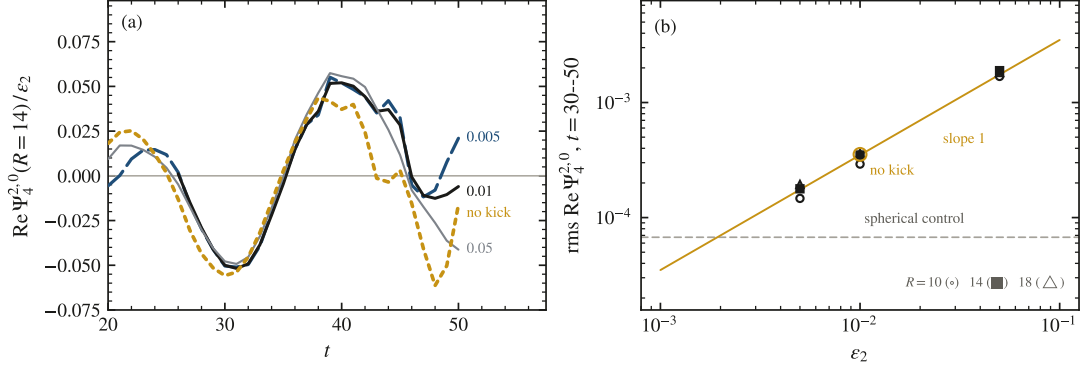


FIG. 11. Linearity of the collapsing throat's wave in its quadrupole seed. (a) $\text{Re } \Psi_4^{2,0}$ at $R = 14$, each arm divided by its seed and named at its end: $\varepsilon_2 = 0.005, 0.01, 0.05$ and the kick-free 0.01 arm collapse onto one curve through the burst; later the smallest arm carries its floor amplified by $1/\varepsilon_2$ and the kick-free arm runs ahead. (b) rms amplitude over $t = 30-50$ against ε_2 at three spheres, with the slope-one line through the 0.01 arm drawn across the decade $10^{-3}-10^{-1}$ as a reference. Open circle: the pure quadrupole, within 2% of the kicked arm. Dashed rule: the floor of the level-4 spherical control (the same kick, no quadrupole) at $R = 14$, which the line crosses at $\varepsilon_2 \approx 2 \times 10^{-3}$. The outcome: the wave is the quadrupole's—linear in ε_2 and unchanged, to 2% in amplitude, when the radial kick is removed. The kick alone radiates only the floor; it moves the wave's phase, not its size.

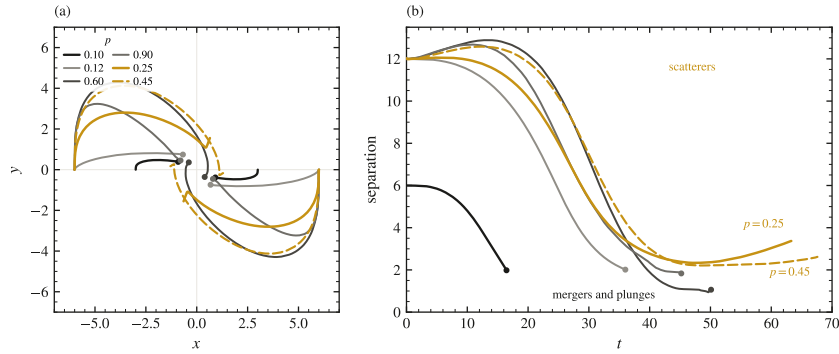


FIG. 12. Merger or fly-by on boosted, solved data: the $d = 12$ momentum scan and the $d = 6$ merger. (a) Both throats in the orbital plane; (b) their separation. The tracks are the coordinate positions of the χ pits, followed through each run by continuity and smoothed over two time units; a track ends (dot) where its pits stop being trackable (a superluminal hop between two χ minima) or at the run's trust window. Gold: the scatterers, $p = 0.25$ (solid, to $t = 63.3$) and 0.45 (dashed, to $t = 67.6$). Ink and greys: the plunges $p = 0.60$ and 0.90 (two legs each, concatenated across their restarts), the $d = 6$, $p = 0.10$ merger, and the $p = 0.12$ arm, whose mouths inflate and whose pits merge near $t = 38$ without a horizon ever forming. Movies: Videos 8 ($p = 0.25$) youtu.be/fh3pU7xCU_I, 9 ($p = 0.60$) youtu.be/-ejFKpetL1Y and 10 ($p = 0.90$) youtu.be/yzvwjC46lJY.

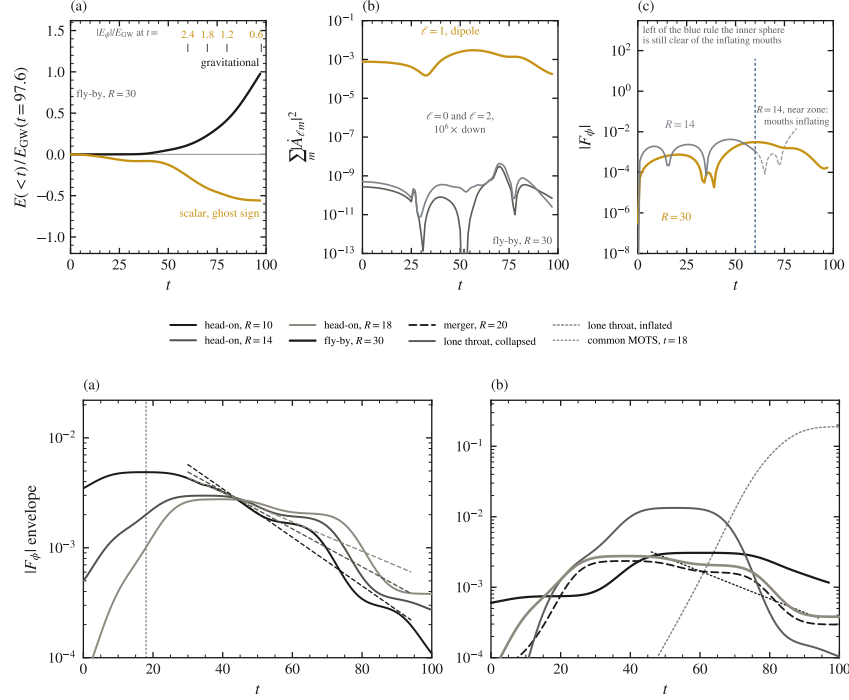


FIG. 13. *Top (channel)*: The scalar channel on the Ψ_4 spheres. (a) Fly-by energy through $R = 30$ in units of the gravitational energy crossed by the end of the sphere’s trusted record ($t = 97.6$, $t - R = 67.6$): gravitational (ink) and scalar with its physical sign (gold, Sec. VIII F). The ratio $|E_\phi|/E_{\text{GW}}$ printed at each cut falls from 2.4 at $t = 60$ to 0.56 over the whole record, as the gravitational burst arrives after the scalar one (Sec. VIII F). (b) Multipole content, fly-by: $\ell = 1$ exceeds $\ell = 0$ and $\ell = 2$ by $10^{6.4}$. (c) Validity window: $|F_\phi|$ at both spheres, each drawn to its retarded gate $t - R = 67.6$. Past the blue rule ($t = 60$) the inner sphere reads the inflating mouths’ near field—its climb there has no counterpart at $R = 30$ one light-crossing later—so that stretch is dashed and carries no claim; the $t = 60$ window at $R = 14$ gives 0.6 in place of 2.4. *Bottom (censorship)*: Envelope of the scalar flux $|F_\phi|$, horizoned against horizonless arms: a running maximum over 25 units (about one dipole period), smoothed in log amplitude, since the raw flux changes sign every ~ 13 units. (a) The head-on chain’s glued record at $R = 10/14/18$. The common MOTS closes at $t = 18$ (dotted); the remaining hair leaves as one $\ell = 1$ pulse, cresting later at each sphere, then decays (dashed: fits over $t = 30\text{--}95$, $\tau = 20/25/32$). The rise before the rule is the open mouths’ static hair, canonically ingoing; only the post-horizon window is integrated. (b) One sphere per fate (head-on and lone throat $R = 18$, merger $R = 20$, fly-by $R = 30$): the merger, behind its horizon from $t = 13$, decays like the head-on ($\tau = 22$, dotted fit); the fly-by, horizonless, shows its burst’s complete arch, drawn to $t - R = 67.6$ (its trust window). Mid-grey: the lone throat twice—the $\varepsilon = +10^{-2}$ level-4 collapse (solid), behind a MOTS from $t = 11$, and the unkicked level-4 throat (dotted), which inflates, horizonless, and grows in the monopole (counted to $t \approx 80$).

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- [1] J. A. Wheeler, “Geons,” *Phys. Rev.* **97**, 511 (1955).
 - [2] J. A. Wheeler, “On the nature of quantum geometrodynamics,” *Ann. Phys. (N.Y.)* **2**, 604 (1957).
 - [3] C. W. Misner and J. A. Wheeler, “Classical physics as geometry,” *Ann. Phys. (N.Y.)* **2**, 525 (1957).
 - [4] T. A. Roman, “Inflating Lorentzian wormholes,” *Phys. Rev. D* **47**, 1370 (1993).
 - [5] J. Garriga, A. Vilenkin, and J. Zhang, “Black holes and the multiverse,” *J. Cosmol. Astropart. Phys.* **02**, 064 (2016).
 - [6] H. Deng, J. Garriga, and A. Vilenkin, “Primordial black hole and wormhole formation by domain walls,” *J. Cosmol. Astropart. Phys.* **04**, 050 (2017).
 - [7] A. A. Kirillov and E. P. Savelova, “Dark matter from a gas of wormholes,” *Phys. Lett. B* **660**, 93 (2008).
 - [8] A. A. Kirillov and E. P. Savelova, “Density perturbations in a gas of wormholes,” *Mon. Not. R. Astron. Soc.* **412**, 1710 (2011).
 - [9] H. G. Ellis, “Ether flow through a drainhole: A particle model in general relativity,” *J. Math. Phys.* **14**, 104 (1973).
 - [10] K. A. Bronnikov, “Scalar-tensor theory and scalar charge,” *Acta Phys. Polon. B* **4**, 251 (1973).
 - [11] M. S. Morris and K. S. Thorne, “Wormholes in spacetime and their use for interstellar travel: A tool for teaching general relativity,” *Am. J. Phys.* **56**, 395 (1988).
 - [12] M. Visser, *Lorentzian Wormholes: From Einstein to Hawking* (AIP Press, Woodbury, NY, 1995).
 - [13] H. Shinkai and S. A. Hayward, “Fate of the first traversible wormhole: Black-hole collapse or inflationary expansion,” *Phys. Rev. D* **66**, 044005 (2002).
 - [14] J. A. González, F. S. Guzmán, and O. Sarbach, “Instability of wormholes supported by a ghost scalar field. II. Nonlinear evolution,” *Class. Quantum Grav.* **26**, 015011 (2009).
 - [15] J. A. González, F. S. Guzmán, and O. Sarbach, “Instability of wormholes supported by a ghost scalar field. I. Linear stability analysis,” *Class. Quantum Grav.* **26**, 015010 (2009).
 - [16] R. Geroch, “Topology in general relativity,” *J. Math. Phys.* **8**, 782 (1967).
 - [17] F. J. Tipler, “Singularities and causality violation,” *Ann. Phys. (N.Y.)* **108**, 1 (1977).
 - [18] S. W. Hawking, “Wormholes in spacetime,” *Phys. Rev. D* **37**, 904 (1988).
 - [19] Ya. B. Zel’dovich and I. D. Novikov, “The hypothesis of cores retarded during expansion and the hot cosmological model,” *Sov. Astron.* **10**, 602 (1967).
 - [20] B. J. Carr and S. W. Hawking, “Black holes in the early Universe,” *Mon. Not. R. Astron. Soc.* **168**, 399 (1974).
 - [21] B. Carr, K. Kohri, Y. Sendouda, and J. Yokoyama, “Constraints on primordial black holes,” *Rep. Prog. Phys.* **84**, 116902 (2021).
 - [22] C. Caprini and D. G. Figueroa, “Cosmological backgrounds of gravitational waves,” *Class. Quantum Grav.* **35**, 163001 (2018).
 - [23] G. Agazie *et al.* (NANOGrav Collaboration), “The NANOGrav 15 yr data set: Evidence for a gravitational-wave background,” *Astrophys. J. Lett.* **951**, L8 (2023).
 - [24] P. Amaro-Seoane *et al.*, “Laser Interferometer Space Antenna,” arXiv:1702.00786 [astro-ph.IM] (2017).
 - [25] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), “Upper limits on the isotropic gravitational-wave background from Advanced LIGO and Advanced Virgo’s third observing run,” *Phys. Rev. D* **104**, 022004 (2021).
 - [26] L. Pagano, L. Salvati, and A. Melchiorri, “New constraints on primordial gravitational waves from Planck 2015,” *Phys. Lett. B* **760**, 823 (2016).
 - [27] B. Allen, W. G. Anderson, P. R. Brady, D. A. Brown, and J. D. E. Creighton, “FINDCHIRP: An algorithm for detection of gravitational waves from inspiraling compact binaries,” *Phys. Rev. D* **85**, 122006 (2012).
 - [28] C. Reisswig and D. Pollney, “Notes on the integration of numerical relativity waveforms,” *Class. Quantum Grav.* **28**, 195015 (2011).
 - [29] S. A. Usman *et al.*, “The PyCBC search for gravitational waves from compact binary coalescence,” *Class. Quantum Grav.* **33**, 215004 (2016).
 - [30] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), “Open data from the third observing run of LIGO, Virgo, KAGRA and GEO,” *Astrophys. J. Suppl.* **267**, 29 (2023).
 - [31] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), “GWTC-3: Compact binary coalescences observed by LIGO and Virgo during the second part of the third observing run,” *Phys. Rev. X* **13**, 041039 (2023).
 - [32] C. W. Misner and D. H. Sharp, “Relativistic equations for adiabatic, spherically symmetric gravitational collapse,” *Phys. Rev.* **136**, B571 (1964).
 - [33] D. Hochberg and M. Visser, “Dynamic wormholes, antitrapped surfaces, and energy conditions,” *Phys. Rev. D* **58**, 044021 (1998).
 - [34] S. W. Hawking, “Gravitational radiation from colliding black holes,” *Phys. Rev. Lett.* **26**, 1344 (1971).
 - [35] D. Alic, C. Bona-Casas, C. Bona, L. Rezzolla, and C. Palenzuela, “Conformal and covariant formulation of the Z4 system with constraint-violating solutions,” *Phys. Rev. D* **85**, 064040 (2012).
 - [36] C. Bona, J. Massó, E. Seidel, and J. Stela, “New formalism for numerical relativity,” *Phys. Rev. Lett.* **75**, 600 (1995).
 - [37] M. Campanelli, C. O. Lousto, P. Marronetti, and Y. Zlochower, “Accurate evolutions of orbiting black-hole binaries without excision,” *Phys. Rev. Lett.* **96**, 111101 (2006).
 - [38] J. G. Baker, J. Centrella, D.-I. Choi, M. Koppitz, and J. van Meter, “Gravitational-wave extraction from an inspiraling configuration of merging black holes,” *Phys. Rev. Lett.* **96**, 111102 (2006).
 - [39] GRTL Collaboration, *GRTeclyn*, <https://github.com/GRTLCollaboration/GRTeclyn>.
 - [40] W. Zhang *et al.*, “AMReX: a framework for block-structured adaptive mesh refinement,” *J. Open Source Softw.* **4**, 1370 (2019).
 - [41] T. Andrade *et al.*, “GRChombo: an adaptable numerical relativity code for fundamental physics,” *J. Open Source Softw.* **6**, 3703 (2021).

- [42] N. M. Shirokov, “Wormhole dynamics: non-linear collapse and gravitational-wave emission,” arXiv:2604.00071 [gr-qc] (2026).
- [43] N. M. Shirokov, “The Bondi dipole in full numerical relativity: a self-accelerating positive-negative mass binary,” arXiv:2608.24577 [gr-qc] (2026).
- [44] Á. Bogdán *et al.*, “Evidence for heavy-seed origin of early supermassive black holes from a $z \approx 10$ X-ray quasar,” *Nat. Astron.* **8**, 126 (2024).
- [45] P. Natarajan, F. Pacucci, A. Ricarte, Á. Bogdán, A. D. Goulding, and N. Cappelluti, “First detection of an overmassive black hole galaxy: UHZ1—evidence for heavy black hole seeds from direct collapse?,” *Astrophys. J. Lett.* **960**, L1 (2024).
- [46] J. E. Greene *et al.*, “UNCOVER spectroscopy confirms the surprising ubiquity of active galactic nuclei in red sources at $z > 5$,” *Astrophys. J.* **964**, 39 (2024).
- [47] J. Matthee *et al.*, “Little red dots: an abundant population of faint active galactic nuclei at $z \sim 5$ revealed by the EIGER and FRESCO JWST surveys,” *Astrophys. J.* **963**, 129 (2024).
- [48] R. Maiolino *et al.*, “A small and vigorous black hole in the early Universe,” *Nature* **627**, 59 (2024).
- [49] E. Bañados *et al.*, “An 800-million-solar-mass black hole in a significantly neutral Universe at a redshift of 7.5,” *Nature* **553**, 473 (2018).
- [50] K. Inayoshi, E. Visbal, and Z. Haiman, “The assembly of the first massive black holes,” *Annu. Rev. Astron. Astrophys.* **58**, 27 (2020).
- [51] T. Nakama, B. Carr, and J. Silk, “Limits on primordial black holes from μ distortions in cosmic microwave background,” *Phys. Rev. D* **97**, 043525 (2018).
- [52] K. Clough, “Continuity equations for general matter: applications in numerical relativity,” *Class. Quantum Grav.* **38**, 167001 (2021).
- [53] E.ourgoulhon, *3+1 Formalism in General Relativity: Bases of Numerical Relativity*, Lecture Notes in Physics Vol. 846 (Springer, Berlin, 2012).
- [54] B. Kleihaus and J. Kunz, “Rotating Ellis wormholes in four dimensions,” *Phys. Rev. D* **90**, 121503(R) (2014).
- [55] V. Dzhunushaliev, V. Folomeev, B. Kleihaus, J. Kunz, and E. Radu, “Rotating wormholes in five dimensions,” *Phys. Rev. D* **88**, 124028 (2013).
- [56] B. Azad, J. L. Blázquez-Salcedo, F. S. Khoo, and J. Kunz, “Are slowly rotating Ellis-Bronnikov wormholes stable?,” *Phys. Lett. B* **848**, 138349 (2024).
- [57] B. Azad, J. L. Blázquez-Salcedo, F. S. Khoo, and J. Kunz, “Radial perturbations of Ellis-Bronnikov wormholes in slow rotation up to second order,” *Phys. Rev. D* **109**, 124051 (2024).
- [58] F. S. Khoo, B. Azad, J. L. Blázquez-Salcedo, L. M. González-Romero, B. Kleihaus, J. Kunz, and F. Navarro-Lérida, “Quasinormal modes of rapidly rotating Ellis-Bronnikov wormholes,” *Phys. Rev. D* **109**, 084013 (2024).
- [59] E. Poisson and M. Visser, “Thin-shell wormholes: Linearization stability,” *Phys. Rev. D* **52**, 7318 (1995).
- [60] P. Kanti, B. Kleihaus, and J. Kunz, “Wormholes in dilaton Einstein-Gauss-Bonnet theory,” *Phys. Rev. Lett.* **107**, 271101 (2011).
- [61] K. A. Bronnikov, J. C. Fabris, and A. Zhidenko, “On the stability of scalar-vacuum space-times,” *Eur. Phys. J. C* **71**, 1791 (2011).
- [62] S. A. Hayward, “Gravitational energy in spherical symmetry,” *Phys. Rev. D* **53**, 1938 (1996).
- [63] E. Babichev, V. Dokuchaev, and Yu. Eroshenko, “Black hole mass decreasing due to phantom energy accretion,” *Phys. Rev. Lett.* **93**, 021102 (2004).
- [64] T. Robson, N. J. Cornish, and C. Liu, “The construction and use of LISA sensitivity curves,” *Class. Quantum Grav.* **36**, 105011 (2019).
- [65] E. Thrane and J. D. Romano, “Sensitivity curves for searches for gravitational-wave backgrounds,” *Phys. Rev. D* **88**, 124032 (2013).
- [66] N. Aghanim *et al.* (Planck Collaboration), “Planck 2018 results. VI. Cosmological parameters,” *Astron. Astrophys.* **641**, A6 (2020).
- [67] A. Ashtekar and B. Krishnan, “Isolated and dynamical horizons and their applications,” *Living Rev. Relativ.* **7**, 10 (2004).
- [68] E. Farhi and A. H. Guth, “An obstacle to creating a universe in the laboratory,” *Phys. Lett. B* **183**, 149 (1987).
- [69] S. K. Blau, E. I. Guendelman, and A. H. Guth, “Dynamics of false-vacuum bubbles,” *Phys. Rev. D* **35**, 1747 (1987).
- [70] A. R. Liddle and S. M. Leach, “How long before the end of inflation were observable perturbations produced?,” *Phys. Rev. D* **68**, 103503 (2003).
- [71] J. M. Cline, S. Jeon, and G. D. Moore, “The phantom menaced: Constraints on low-energy effective ghosts,” *Phys. Rev. D* **70**, 043543 (2004).
- [72] S. A. Hayward, “General laws of black-hole dynamics,” *Phys. Rev. D* **49**, 6467 (1994).
- [73] S. A. Hayward, “Dynamic wormholes,” *Int. J. Mod. Phys. D* **8**, 373 (1999).
- [74] C. Gundlach, “Pseudospectral apparent horizon finders: An efficient new algorithm,” *Phys. Rev. D* **57**, 863 (1998).
- [75] M. Alcubierre *et al.*, “Gauge conditions for long-term numerical black hole evolutions without excision,” *Phys. Rev. D* **67**, 084023 (2003).
- [76] G. Huisken and T. Ilmanen, “The inverse mean curvature flow and the Riemannian Penrose inequality,” *J. Differ. Geom.* **59**, 353 (2001).
- [77] M. Mars, “Present status of the Penrose inequality,” *Class. Quantum Grav.* **26**, 193001 (2009).
- [78] T. Damour, F. Guercilena, I. Hinder, S. Hopper, A. Nagar, and L. Rezzolla, “Strong-field scattering of two black holes: Numerics versus analytics,” *Phys. Rev. D* **89**, 081503(R) (2014).
- [79] E. Berti, V. Cardoso, and C. M. Will, “Gravitational-wave spectroscopy of massive black holes with the space interferometer LISA,” *Phys. Rev. D* **73**, 064030 (2006).
- [80] E. Berti, V. Cardoso, and A. O. Starinets, “Quasinormal modes of black holes and black branes,” *Class. Quantum Grav.* **26**, 163001 (2009).
- [81] M. A. Scheel, M. Boyle, T. Chu, L. E. Kidder, K. D. Matthews, and H. P. Pfeiffer, “High-accuracy waveforms for binary black hole inspiral, merger, and ringdown,” *Phys. Rev. D* **79**, 024003 (2009).
- [82] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), “All-sky search for short gravitational-wave bursts in the third Advanced LIGO and Advanced Virgo run,” *Phys. Rev. D* **104**, 122004 (2021).
- [83] C. Gao, X. Chen, V. Faraoni, and Y.-G. Shen, “Does the mass of a black hole decrease due to the accretion

- of phantom energy?," *Phys. Rev. D* **78**, 024008 (2008).
- [84] A. Doroshkevich, J. Hansen, I. Novikov, and A. Shatskiy, "Passage of radiation through wormholes," *Int. J. Mod. Phys. D* **18**, 1665 (2009).
 - [85] F. Cremona, F. Pirotta, and L. Pizzocchero, "On the linear instability of the Ellis–Bronnikov–Morris–Thorne wormhole," *Gen. Relativ. Gravit.* **51**, 19 (2019).
 - [86] S. A. Hayward, "Wormhole dynamics in spherical symmetry," *Phys. Rev. D* **79**, 124001 (2009).
 - [87] H. Witek, M. Zilhão, L. Gualtieri, V. Cardoso, C. Herdeiro, A. Nerozzi, and U. Sperhake, "Numerical relativity for D dimensional space-times: Head-on collisions of black holes and gravitational wave extraction," *Phys. Rev. D* **82**, 104014 (2010).
 - [88] I. D. Novikov and D. I. Novikov, "Collapse of a wormhole and its transformation into black holes," *J. Exp. Theor. Phys.* **129**, 495 (2019).
 - [89] H. Deng and A. Vilenkin, "Primordial black hole formation by vacuum bubbles," *J. Cosmol. Astropart. Phys.* **12**, 044 (2017).
 - [90] Y. Gouttenoire and E. Vitagliano, "Primordial black holes and wormholes from domain wall networks," *Phys. Rev. D* **109**, 123507 (2024).
 - [91] N. S. Kardashev, I. D. Novikov, and A. A. Shatskiy, "Astrophysics of wormholes," *Int. J. Mod. Phys. D* **16**, 909 (2007).
 - [92] C. Bambi, "Can the supermassive objects at the centers of galaxies be traversable wormholes? The first test of strong gravity for mm/sub-mm very long baseline interferometry facilities," *Phys. Rev. D* **87**, 107501 (2013).
 - [93] C. Gundlach and J. M. Martín-García, "Critical phenomena in gravitational collapse," *Living Rev. Relativ.* **10**, 5 (2007).
 - [94] M. W. Choptuik, T. Chmaj, and P. Bizoń, "Critical behavior in gravitational collapse of a Yang-Mills field," *Phys. Rev. Lett.* **77**, 424 (1996).
 - [95] R. Penrose, "Gravitational collapse: The role of general relativity," *Riv. Nuovo Cimento* **1**, 252 (1969) [*Gen. Relativ. Gravit.* **34**, 1141 (2002)].
 - [96] R. M. Wald, "Gravitational collapse and cosmic censorship," in *Black Holes, Gravitational Radiation and the Universe*, edited by B. R. Iyer and B. Bhawal (Springer, Dordrecht, 1999), p. 69.
 - [97] S. M. Carroll, M. Hoffman, and M. Trodden, "Can the dark energy equation-of-state parameter w be less than -1 ?" *Phys. Rev. D* **68**, 023509 (2003).
 - [98] V. Cardoso, E. Franzin, and P. Pani, "Is the gravitational-wave ringdown a probe of the event horizon?," *Phys. Rev. Lett.* **116**, 171101 (2016); **117**, 089902(E) (2016).
 - [99] P. Bueno, P. A. Cano, F. Goelen, T. Hertog, and B. Vernocke, "Echoes of Kerr-like wormholes," *Phys. Rev. D* **97**, 024040 (2018).
 - [100] J. L. Friedman, K. Schleich, and D. M. Witt, "Topological censorship," *Phys. Rev. Lett.* **71**, 1486 (1993); **75**, 1872(E) (1995).
 - [101] D. Christodoulou, "Nonlinear nature of gravitation and gravitational-wave experiments," *Phys. Rev. Lett.* **67**, 1486 (1991).
 - [102] K. S. Thorne, "Gravitational-wave bursts with memory: The Christodoulou effect," *Phys. Rev. D* **45**, 520 (1992).
 - [103] M. Favata, "Post-Newtonian corrections to the gravitational-wave memory for quasi-circular, inspiralling compact binaries," *Phys. Rev. D* **80**, 024002 (2009).